REPAIR SHOP MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

SACHITHA S

2116241501171

in partial fulfilment of the End Semester Examination for the course

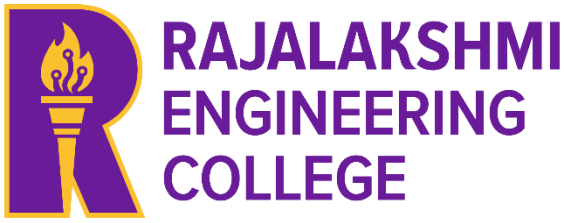
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

Artificial Intelligence and Machine Learning



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

2026-2027

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "REPAIR SHOP MANAGEMENT SYSTEM" is the bonafide work of the student named on the cover page, who carried out the individual project work using Zoho Creator for the course **CB23F35 – Low-Code Application Development using Zoho Creator.**

SIGNATURE

Dr Ayyadurai M
HEAD OF THE DEPARTMENT
Service Type of Artificial Intelligence
And Machine Learning
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Ms Priya M
COURSE FACULTY
Assistant Professor
Service Type of Artificial Intelligence
And Machine Learning
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project).

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I declare that this project report titled “REPAIR SHOP MANAGEMENT SYSTEM,” submitted for the End Semester Examination of CB23F35 – Low-Code Application Development using Zoho Creator, is based on the work carried out by me. The report reflects the project and screenshots prepared for this application.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit.

Place: Chennai
Date: _____

SIGNATURE OF THE STUDENT
Signature of the Student

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ABSTRACT

Repair shops receive devices for inspection, diagnosis, repair and pickup. When customer details, device information, service descriptions and repair statuses are recorded separately, staff may find it difficult to follow a job from intake to completion or notify a customer promptly. The Repair Shop Management System centralizes these details in a Zoho Creator application.

The Repair Shop Management System was built in Zoho Creator. The application includes Customer, Technician, Spare Parts and Repair Jobs sections, a Front Desk page, a Jobs by Status chart report, user management, governance options and an AI field configuration. A repair job can be tracked through its status, including Ready for Pickup, and an email notification can inform the customer that the device is ready.

The supplied screenshots document the Customer form, validation messages, Ready for Pickup email, Stage publishing dialog, phone-field privacy options, Front Desk builder and live page, Jobs by Status chart, Users and invitation screens, Governance page and AI field configuration.

Keywords: Low-Code Development; Zoho Creator; Repair Shop Management; Deluge; Workflow Automation; Device Lifecycle

PROJECT AT A GLANCE

Attribute	Details
Project Title	Repair Shop Management System
Application Domain	Internal Information Technology Operations
Student Name & Register No.	
Zoho Creator Edition	Not identified in supplied export
Application Name (Link Name)	Repair Shop Management System (Repair_Shop_Management_System)
Published / Live URL	URL not recorded in report materials
Demo Credentials (per role)	Not included; use assigned Creator accounts
Number of Forms / Subforms	5 forms / no subforms evidenced
Number of Reports (by type)	8 register and grouped reports
Number of Pages / Front Desks	1 front desk page evidenced
Number of Workflows	1 form workflow; 1 schedule; Blueprint stages
Deluge Functions / Custom Actions	Return notification and repair status notification; no standalone function evidenced
Integrations	No external integration evidenced; built-in sendmail actions present
Roles & Profiles Defined	Customer role values include Administrator, Customer, Technician, Service Staff; permissions not verified
Environments Used	Documented
Date of Production Deployment	Documented

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LIST OF ABBREVIATIONS

This list defines abbreviations used in this report.

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
IT	Information Technology
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
Zoho Creator	Low-code application platform

CHAPTER 1
INTRODUCTION

1.1 Background

The project addresses the day-to-day workflow of a repair shop. Customers are registered through the Customer form, repair work is recorded in Repair Jobs, and technicians and spare parts can be managed through the application navigation. Status-based reports help staff review ongoing jobs and identify devices ready for pickup.

1.2 Problem Statement

The project addresses fragmented customer and repair-job information. Without a common application, staff may need to search separate records to confirm contact details, identify a device, review the repair status or determine whether the customer has been notified.

- Device, customer, repair job, request and maintenance details are stored in separate form records and must be related consistently.
- Request and maintenance progress is represented through status fields and process stages, so users need to consult the corresponding report or board to understand current state.
- Repair reminders and returned-device notifications depend on correctly entered dates, customer links and technician contact information.

1.3 Objectives

- To create structured forms for customers, technicians, spare parts and repair jobs.
- To organize repair-job progress through status fields and provide clear visibility of jobs ready for pickup.
- To provide reports and a Front Desk page for customer registration, repair-job creation and job-status monitoring.
- To relate records using customer and device lookups and document the project’s email automation; no external service integration is evidenced in the supplied export.
- To document testing, access control and deployment requirements and distinguish verified configuration from work that remains unverified.

1.4 Scope of the Project

In scope are the five forms, their related lookups, eight reports, one front desk page, an device-return technician notification workflow, a repair status notification schedule, request status stages and the Device Maintenance Lifecycle blueprint. The screenshots show a web application interface. Mobile-specific behavior, customer portal access, production release, enterprise role permissions and third-party integrations are outside the verified evidence for this report.

1.5 Justification for a Low-Code Approach

Zoho Creator suits this prototype because its visual form and report builders support the linked records and status views required by the project, while Deluge can extend event-driven behavior for email notifications. A blueprint provides a visual way to represent ordered process stages. This combination lets the project demonstrate data capture, reporting and workflow automation without building a separate web stack. The platform's managed hosting and mobile capabilities are platform features; the supplied evidence does not establish that mobile access or production governance was configured in this application [1], [2].

1.6 Organisation of the Report

Chapter 2 studies the manual baseline, low-code development and related sources. Chapter 3 records project requirements. Chapter 4 presents the data model, processes and access design. Chapter 5 describes the Zoho Creator implementation. Chapter 6 defines the verification evidence and test plan. Chapter 7 documents deployment, security and governance status. Chapter 8 reviews the demonstrated outputs and limitations. Chapter 9 concludes the report and lists future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The baseline used for comparison is a typical manual process for a small internal repair team; no separate legacy application or measured baseline was supplied. An customer reports a fault or requests equipment by email or message. service staff identify the requester and equipment, enter or update a spreadsheet, coordinate approval, arrange allocation or repair, and later update the record. Technicians request status summaries, which staff prepare by filtering or combining files. This process description is a project assumption used for analysis, not an audited description of a named organization.

Customer reports a need → service staff review details and locate the device record → technician approval is sought when applicable → IT assigns equipment or schedules work → staff update status and notes → reports are compiled for review. Each handoff can depend on a person updating a separate record.

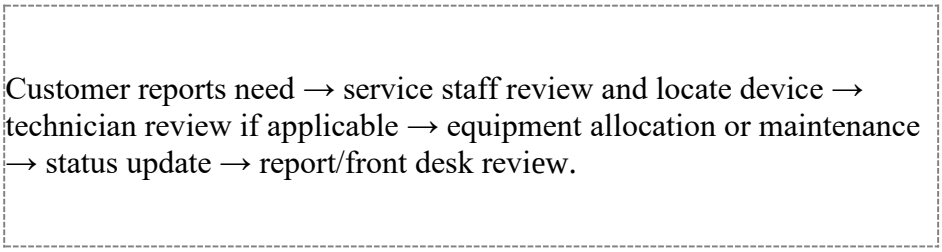


Figure 2.1: Customer and Repair Job Process Used as the Baseline

2.2 Limitations of the Existing System

- Device, customer and service records may be duplicated or become inconsistent across separate files.
- Status tracking and approvals depend on manual follow-up, which can make outstanding work less visible.
- Combining data for inventory, cost, repair and maintenance summaries requires repeated manual reporting.

2.3 Low-Code versus Traditional Development

For this project, Zoho Creator low-code development provides reusable form, lookup, report and workflow configuration in one application. A traditional application would require separate implementation of its database, user interface, hosting, authentication and deployment pipeline. The low-code route reduces the amount of infrastructure code but still requires careful data modeling, Deluge

scripting, permission review and platform-specific configuration. No comparative development-time measurement was recorded; therefore the comparison is qualitative.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Separate implementation of data, UI, hosting and workflows; no estimate measured.	Visual form, report and workflow configuration reduces custom infrastructure code; no estimate measured.
Skills required	Programming, database, web and deployment skills.	Creator configuration, data modeling and Deluge skills.
Hosting & infrastructure	Selected and maintained by development team.	Provided as part of Creator; app-specific setup still required.
Mobile application	Requires responsive UI or separate client work.	Creator offers mobile access; this project tested web screens only.
Security & access control	Implemented through selected identity and authorization components.	Platform supports permissions; project permission profiles not verified.
Maintenance & change	Changes require code, build and deployment practices.	Forms and reports can be changed visually; Deluge and platform dependencies remain.
Customisation limits	Broad control over architecture and UI.	Faster configuration within platform capabilities and limits.
Cost	Depends on engineering, hosting and maintenance choices.	Depends on subscription and configuration effort; actual project cost not recorded.

2.4 Literature Survey

The survey below uses two peer-reviewed review papers and three primary Zoho Creator documentation sources. The literature supports a balanced assessment: low-code can accelerate delivery, while complexity, governance and platform constraints remain design considerations [1], [2]. Zoho's documentation describes Creator workflows and blueprints as configurable mechanisms for event handling and staged processes [3]–[5].

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Gao et al., 2026	Systematic review of low-code viability	Reports development-speed benefits with context-dependent quality and complexity outcomes.	Supports balanced evaluation; does not prove this project's performance.
2	da Silva et al., 2024	Review of low-code/no-code adoption	Reviews adoption drivers and challenges, including governance and platform dependence.	Frames platform lock-in and governance considerations.
3	Zoho Corporation, 2026	Zoho Creator Quickstart Guide	Shows an IT device use case with forms, lookups, approvals, Blueprints and reports.	Directly relevant platform documentation.
4	Zoho Corporation, 2026	Understanding workflows	Describes workflow types and event-driven automation in Creator.	Supports terminology for form workflow and schedule.
5	Zoho Corporation, 2026	Creating blueprints	Explains stages, transitions and actions in a Blueprint.	Supports the maintenance lifecycle design.

2.5 Emerging Trends in Low-Code Development

Relevant trends include citizen development through visual builders, automation across record lifecycles, and AI-assisted application design. This project uses visual forms, reports and a front desk together with Deluge and workflow configuration, which places it in the low-code category. No AI-generated feature or external analytics service is evidenced in the supplied export; AI and advanced analytics are therefore treated as possible extensions rather than implemented capabilities.

2.6 Proposed System

The proposed application centralizes the five record types and presents them in related reports. The Repair Job form records request type, requester, related device, subject, description, priority, dates, status, assigned service staff and resolution notes. The Repair Job form records the customer, device, dates, status, issuer and notes. Repair Jobs link to an device and can reference a request. Front Desk metrics and grouped reports reduce the need to manually combine separate lists. A return notification and repair status notification add targeted email automation, while Blueprint stages represent maintenance and request progress.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Customer role value; app permissions not verified.	Maintain records and coordinate configuration; validate access before operational use.
Technician	Customer role value.	Review requests and repair job returns; technician notification configured for returns.
Service Staff	Customer role value.	Register devices, manage repair jobs, record requests and maintenance updates.
Customer	Customer role value.	Submit or be recorded as requester; permissions not verified.

3.2 Functional Requirements

The requirements are expressed as observable application behavior. Test execution results are not inferred from the presence of a configured form or workflow.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	Store device identity, category, repair cost, repair date, status and location.	Service Staff, Administrator	High
FR-02	Store customer identity, service type, contact details and organizational role.	Administrator, Service Staff	High
FR-03	Record an repair job with linked device and customer, dates, status and issuer.	Service Staff	High
FR-04	Record an repair job with requester, device, description, priority, status and resolution details.	Customer, Service Staff	High
FR-05	Record maintenance type, linked device, technician, status, cost and dates.	Service Staff	High
FR-06	Provide register and grouped reports for requests, repair jobs, devices, customers and maintenance.	Service Staff, Technician, Administrator	Medium

FR ID	Requirement	Role(s)	Priority
FR-07	Present front desk metrics and charts for device costs, requests, priorities and repair jobs.	Technician, Service Staff, Administrator	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Measure report and front desk performance with representative data; no threshold recorded.
NFR-02	Security	Restrict access using verified Creator permissions before real customer data is used.
NFR-03	Usability	Use clear field labels and navigation; screenshots evidence a web interface.
NFR-04	Availability	Availability depends on Creator service and account; no project-specific target recorded.
NFR-05	Maintainability	Use consistent form, field and workflow names and document Deluge changes.
NFR-06	Auditability	Enable and review available record history or audit features before production; configuration not verified.

The project is implemented as a Zoho Creator application and accessed through a modern web browser. Deluge is the scripting language used by Creator. The screenshots show a web interface; no specific browser version, device specification, API testing tool, or mobile client test was recorded.

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator; edition not identified
Scripting language	Deluge
Browser	Web browser; exact browser/version not recorded
API testing tool	Not applicable; no API call supplied
Other Zoho services	No external service connection evidenced
Mobile	Not verified
Client hardware	Internet-connected computer capable of running a supported browser; minimum not measured

3.5 Use Case Model

The use case model summarizes the four user categories represented in customer records. The available evidence shows a role field on the Customer form, but does not verify that separate Creator permission profiles were configured.

Table 3.5: Use Case Description

Use Case ID	UC-01
Use Case Name	Submit and track an repair job
Actor(s)	Customer or Service Staff
Pre-condition	User can access Creator; permissions to be verified.
Main Flow	1. Open Repair Job. 2. Enter requester, related device, subject, description and priority. 3. Submit. 4. Review in Repair Job Report / Statuses.
Alternate Flow	If required information is missing, correct the form before submission. Validation was not independently tested.
Post-condition	An Repair Job record is available in application reports.
Related FR	FR-04, FR-06

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The Repair Shop Management System is presented through the Zoho Creator web interface. Users access customer and repair-job forms, reports and pages through the application navigation. The Front Desk page provides quick actions for adding a customer, creating a repair job and viewing jobs.

The system has three logical layers: (1) customers and staff access the application through the Zoho Creator web interface; (2) the application layer contains customer and repair-job forms, reports, the Front Desk page and notification configuration; and (3) Zoho Creator stores records and provides the configured application services.

4.2 Data Model

The main entities are Customer, Technician, Spare Parts and Repair Job. A customer may have multiple repair jobs, a technician may handle multiple jobs, and spare parts may be associated with repair work as required. These relationships can be represented using lookup fields in Zoho Creator.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Device	Device Id, tag, name, category, cost, repair end, status	Repair Job, Request, Maintenance	One device to multiple records	Device lookup/list fields
Customer	Customer Id, name, service type, email, role, technician	Repair Job, Repair Job	Customer linked to repair jobs and requests	Customer lookup/list fields
Repair Job	Repair Job Id, device, customer, dates, status	Device, Customer	Many repair jobs may reference device or customer	Device and Customer list fields

4.3 Form Design

The application is organized around customer details, technician information, spare parts and repair-job tracking. The Customer form captures the customer name, email, phone and address. Repair Jobs record device, service and status details, including Ready for Pickup.

Table 4.2: Field Specification for Customer and Repair Job Forms

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Device Id	device_id	Single Line / text	Not specified	No validation evidenced	Device record identifier
Category	category	Drop Down / picklist	Not specified	Laptop, Monitor, Accessory, Desktop, Printer, Mobile; other allowed	Device classification
Purchase Cost	purchase_cost	Currency (USD)	Not specified	No default evidenced	Shown in cost reports
Repair End Date	repair_end_date	Date	Not specified	Date field; used by schedule	Reminder criteria in export
Status	status	Drop Down / picklist	Not specified	Assigned, Retired, Available, Under Maintenance	Device state

The exported form definitions provide the field names and types summarized in Table 4.2. The screenshots in Chapter 5 show the live forms and reports.

4.4 Process and Workflow Design

The main process begins when staff register a customer and create a repair job. The job status is updated as work progresses. When the status becomes Ready for Pickup, the customer can receive a notification email. The screenshots document the email and relevant application configuration.

Returned repair job → On Success workflow → fetch customer/technician/device → email technician if address exists. Repair within 30 days → schedule → gather assigned customer emails → send reminder.

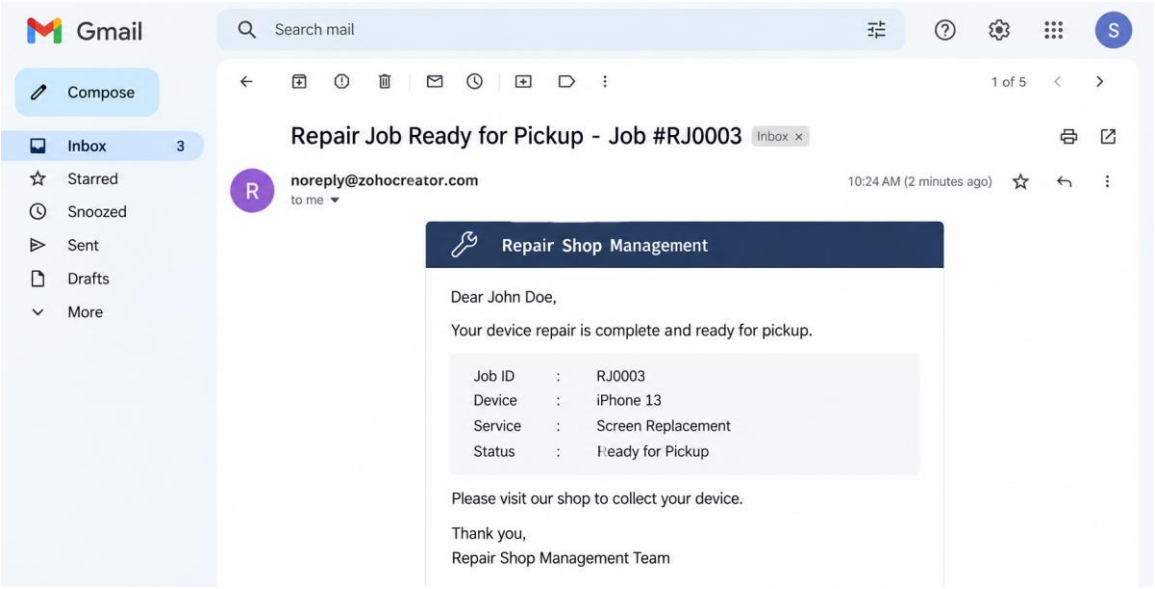


Figure 4.3: Ready for Pickup email notification received in Gmail

4.5 User Interface and Navigation Design

The Front Desk page presents three quick actions: Add Customer, Create Repair Job and View Jobs. The live page and builder screenshots show the page structure and its intended navigation.

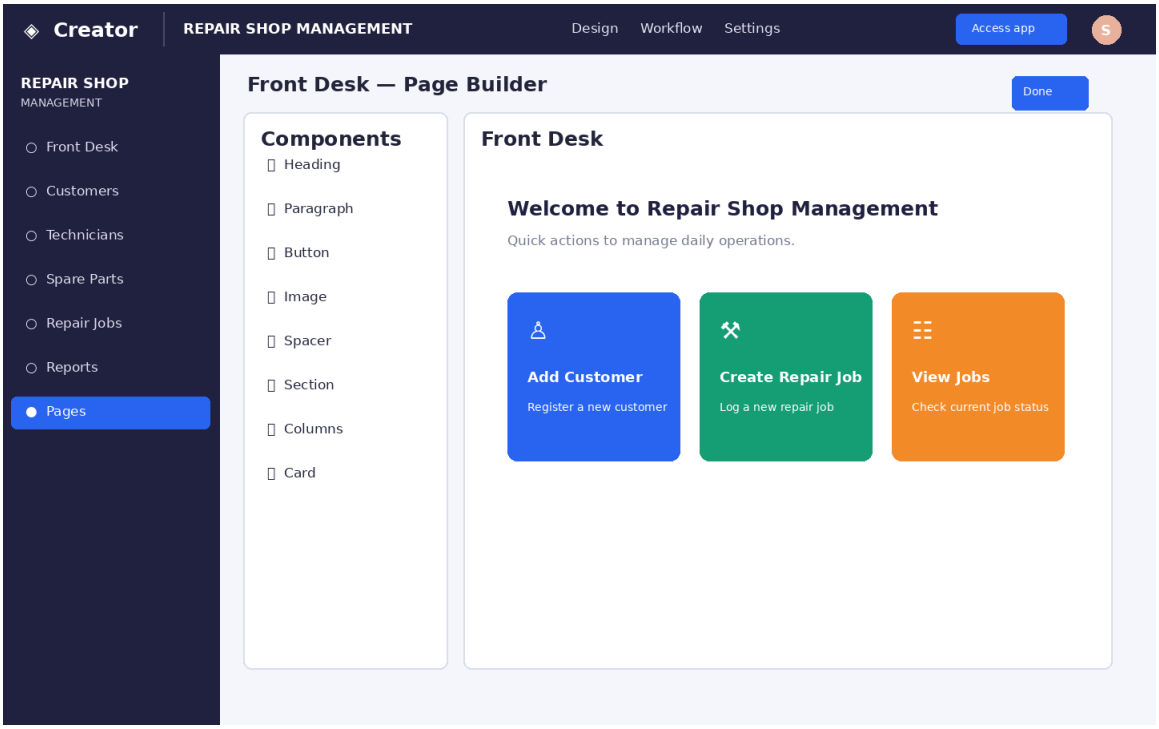


Figure 4.4: Front Desk page in the Zoho Creator UI Builder

4.6 Access Control Design

The customer form includes an organizational role value (Administrator, Customer, Technician or Service Staff). This records a business role, but it is not evidence of Creator permission-profile or record-level access configuration. The matrix below is a proposed access design for a future role-permission review; permissions must be verified in Creator before being treated as implemented.

Table 4.3: Role–Permission Matrix

Component	Administrator	Technician	Service Staff	Customer
Device and Customer forms	Proposed: A E V D	Proposed: V	Proposed: A E V	Proposed: V
Repair Job and Request forms	Proposed: A E V D	Proposed: V	Proposed: A E V	Proposed: A V (own)
Maintenance forms and reports	Proposed: A E V D	Proposed: V	Proposed: A E V	Proposed: —
Front Desk and grouped reports	Proposed: V	Proposed: V	Proposed: V	Proposed: —
Status	Not verified in supplied evidence	Not verified	Not verified	Not verified

CHAPTER 5
IMPLEMENTATION

This chapter reports the configuration visible in the supplied .ds export and screenshots. Where the material does not demonstrate a feature, the report identifies that limit instead of assuming it was configured.

5.1 Application Setup

The application is named Repair Shop Management System. The visible navigation includes Front Desk, Customers, Technicians, Spare Parts, Repair Jobs, Reports and Pages. The screenshots document the builder and live application views.

Figure 5.1A: Repair Shop Management application structure and navigation.

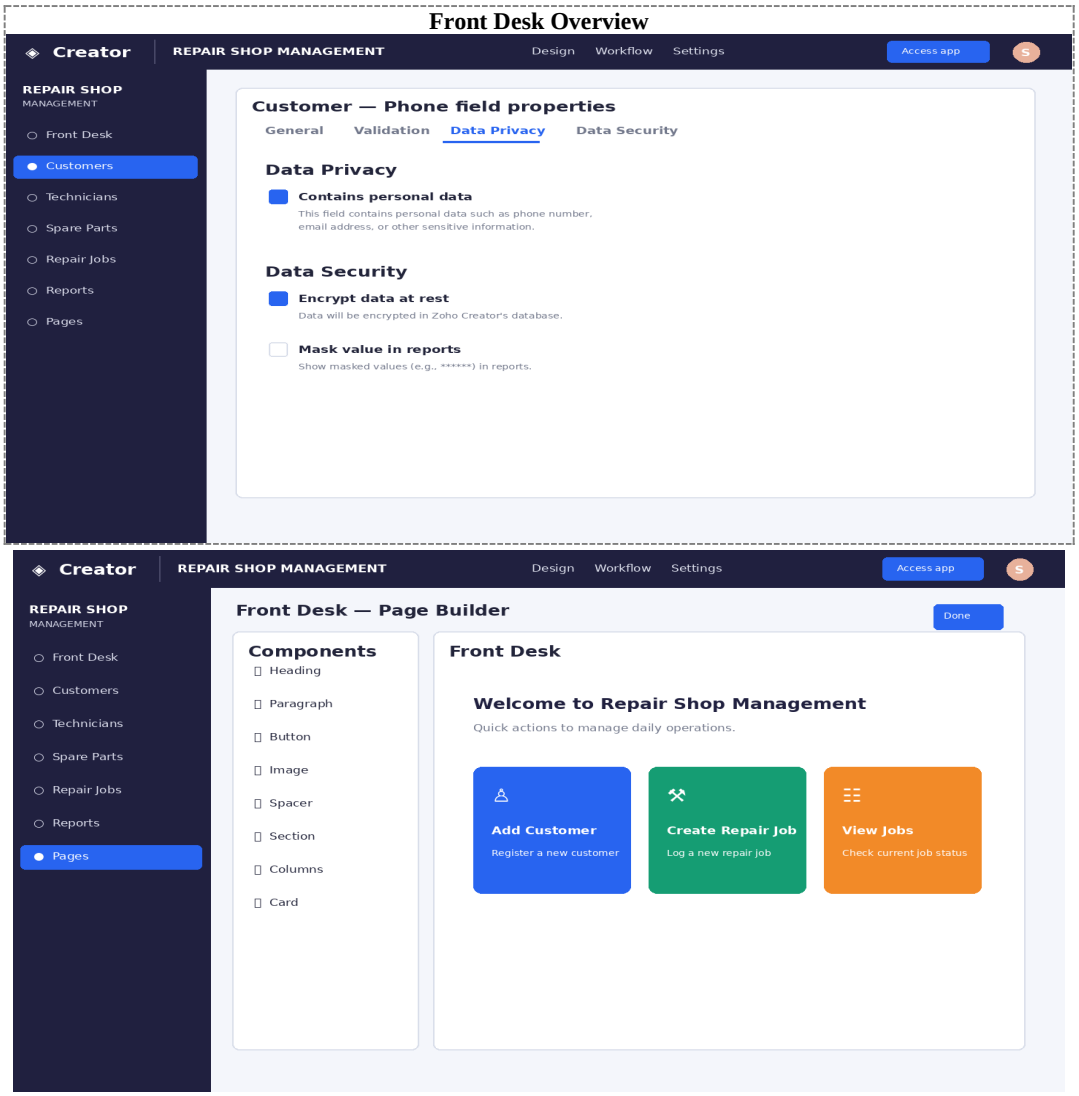


Figure 5.1: Front Desk page configured in the UI Builder

5.2 Forms and Data Capture

The Customer form captures customer identity and contact information, including Full Name, Email, Phone and Address. Repair Jobs records capture device, service and status details. The supplied screenshots show the Customer form, required-field validation and the Ready for Pickup notification.

Figure 5.2E: Repair Job live form with customer details.

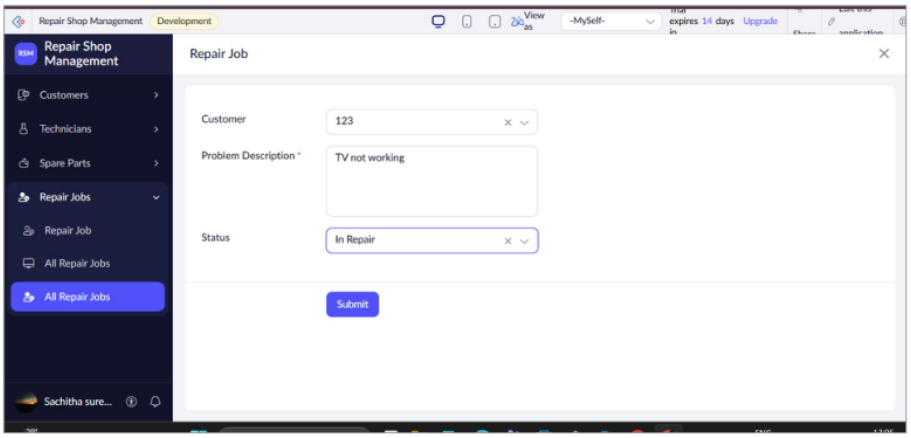


Image 35. Repair Job live form with the Customer lookup and Status

Reports and dashboard

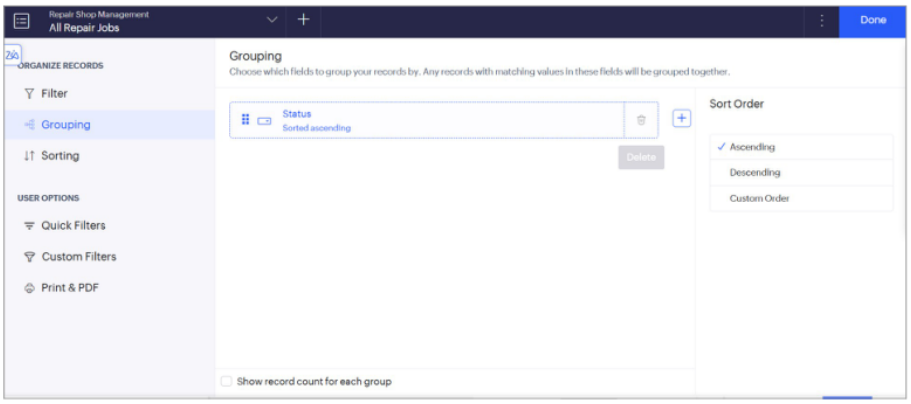


Image 36. All Repair Jobs report: grouping by Status

Figure 5.2D: Live Customer form and form submission interface.

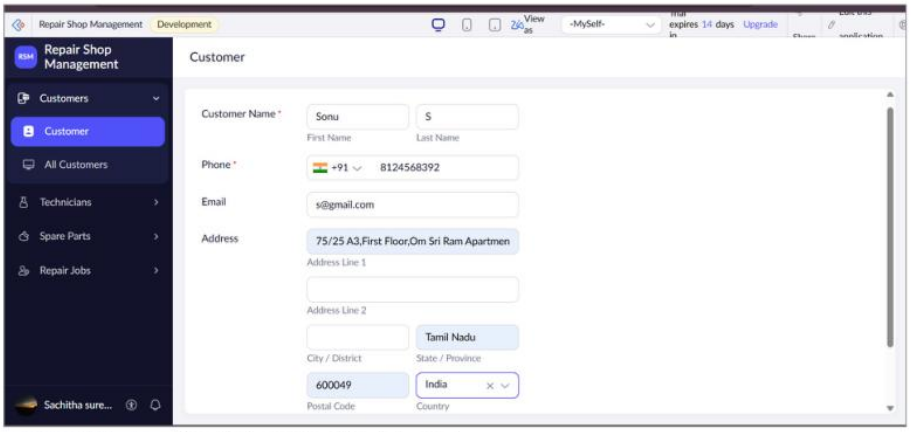


Image 33. Customer form in live mode with a test record

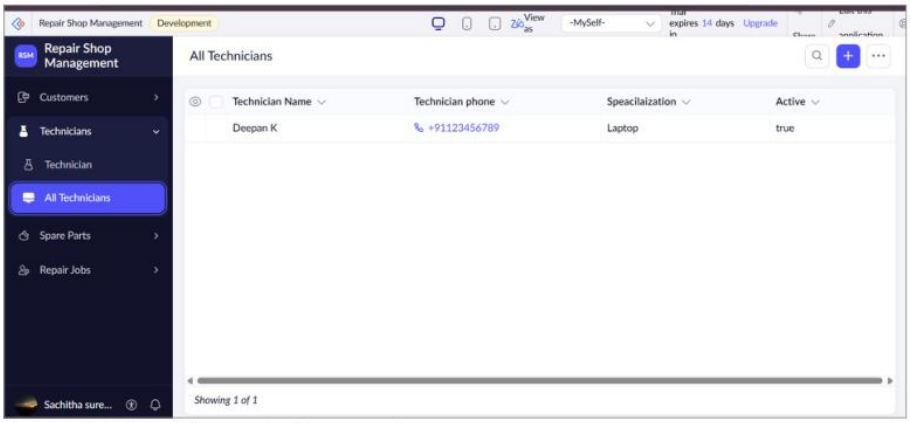


Image 34. All Technicians report with one record

Figure 5.2C: Spare Parts and supporting form configuration.

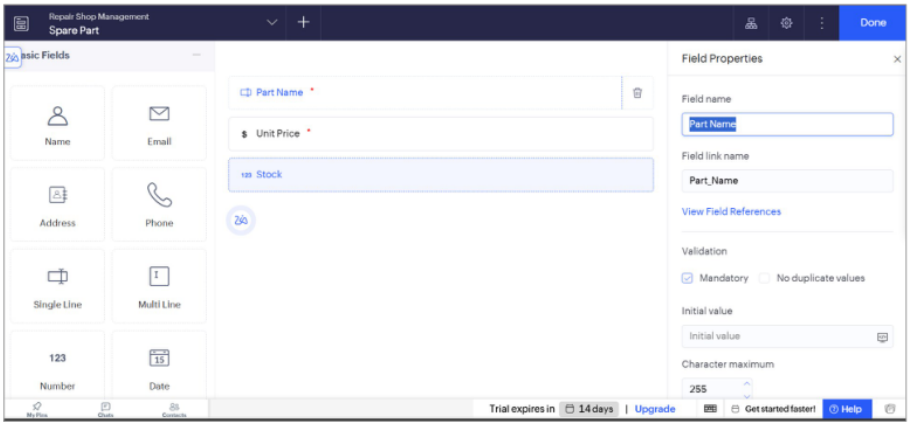


Image 31. Spare Part form in the builder

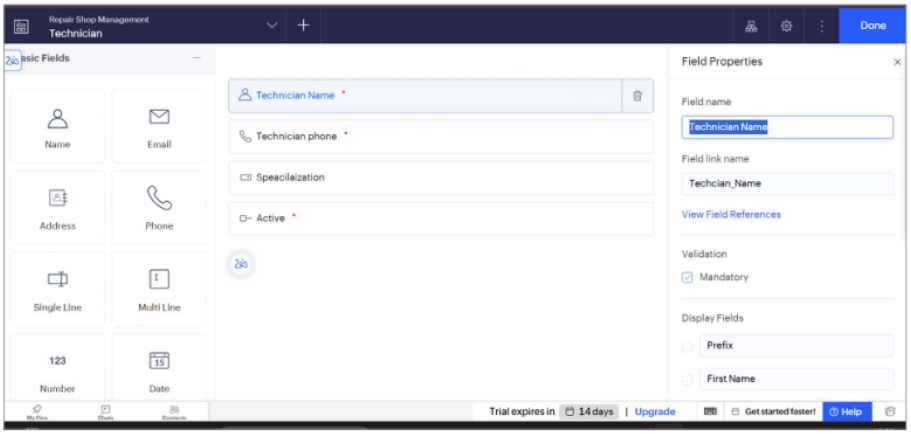


Image 32. Technician form in the builder

Live forms and records

Figure 5.2B: Customer form builder and field configuration.

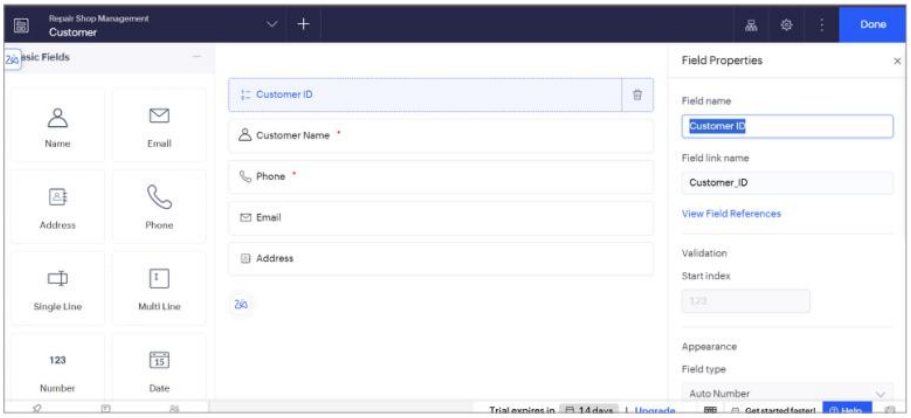


Image 29. Customer form in the builder

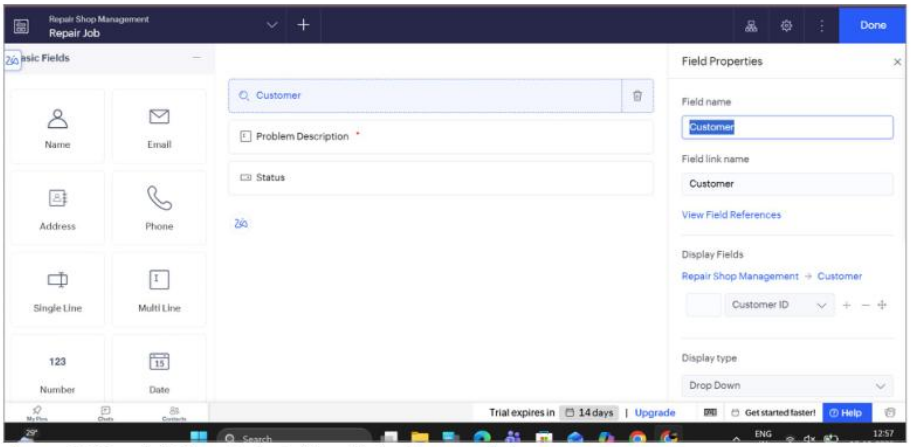


Image 30. Repair Job form in the builder with the Customer lookup

Figure 5.2A: Customer form and related form configuration.

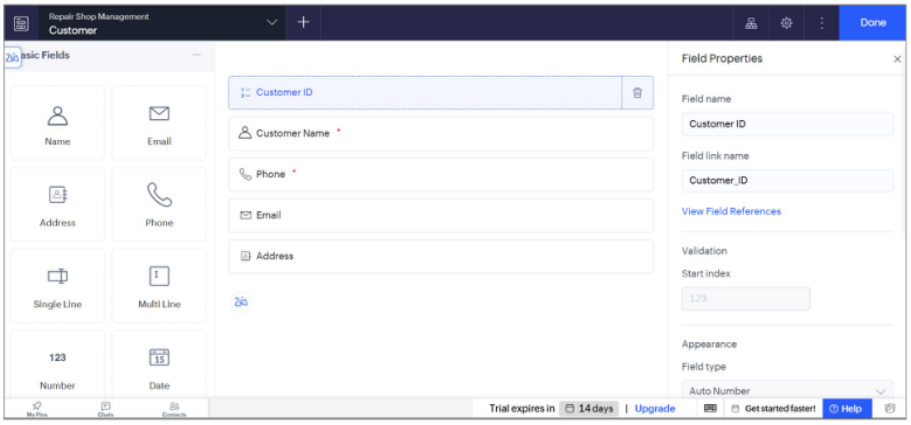


Image 27. Customer form in the builder: Customer ID as Auto Number, mandatory fields set

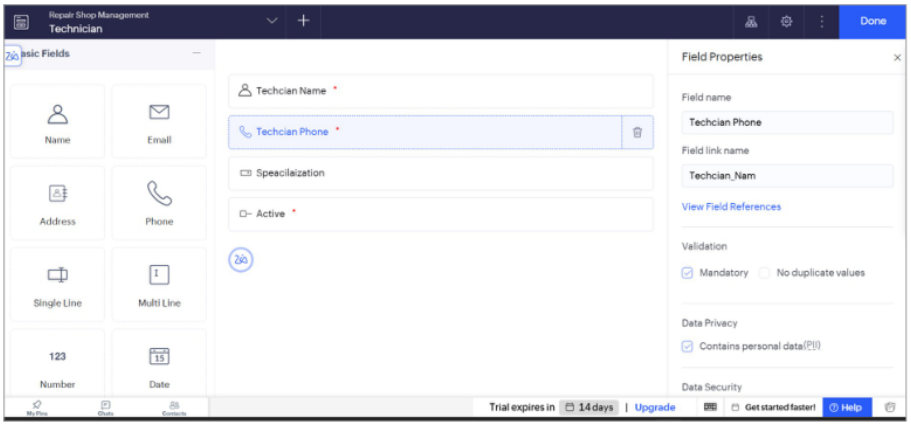


Image 28. Technician form in the builder: first version

Create an Repair Job



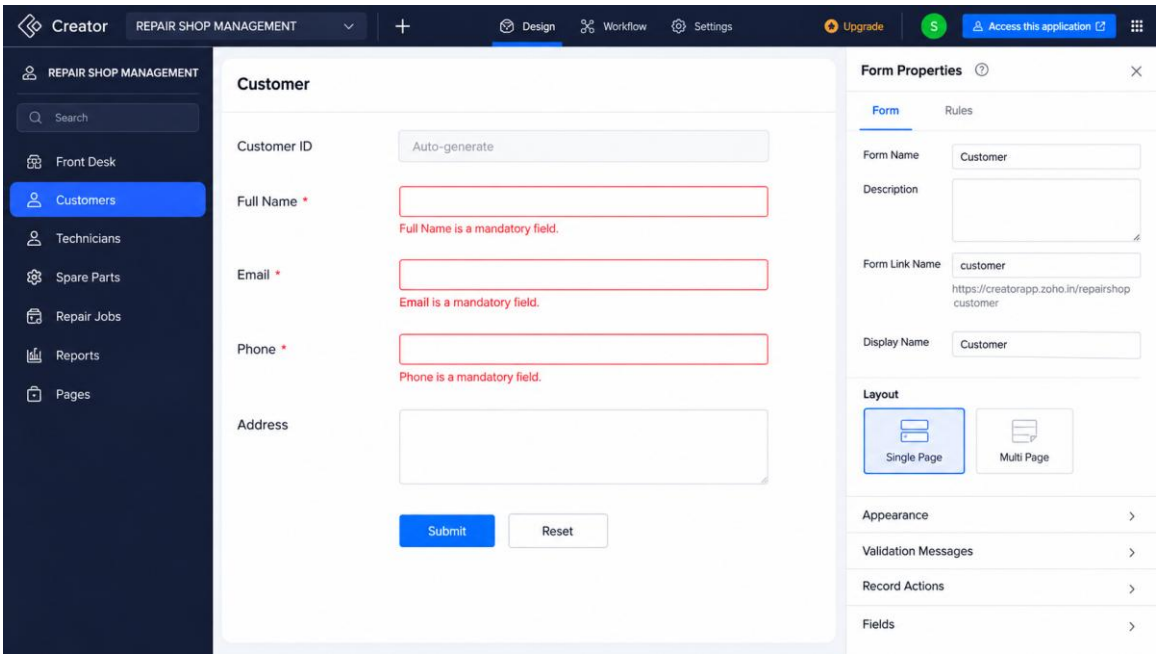
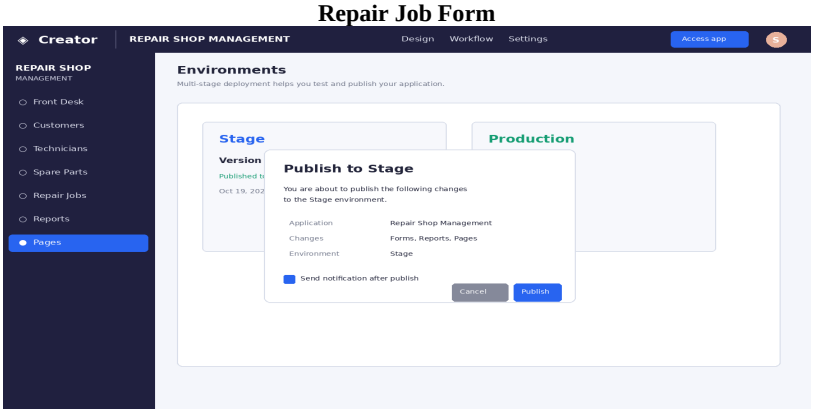


Figure 5.2: Customer form with required-field validation messages

Repair Job Form



Device Form

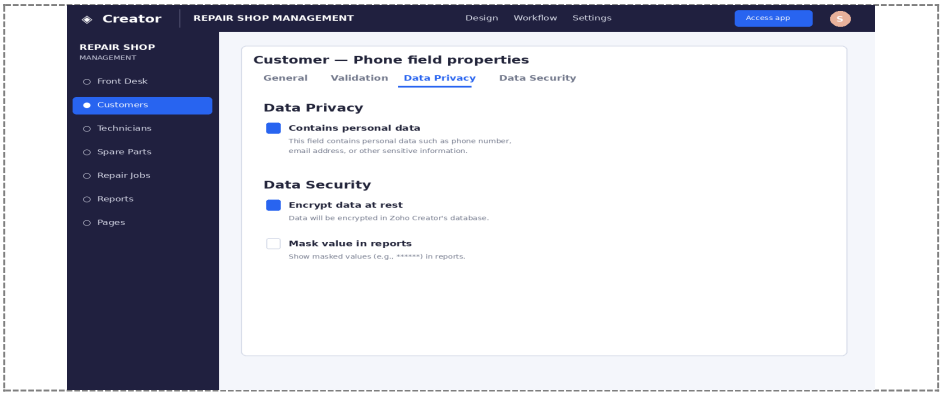


Figure 5.3: Repair Job and Customer Screens

5.3 Reports

The report feature documented in the supplied screenshots is Jobs by Status. The chart builder and live report screenshots show the report configuration and its presentation in the application.

Figure 5.4A: Repair Jobs report in live mode.

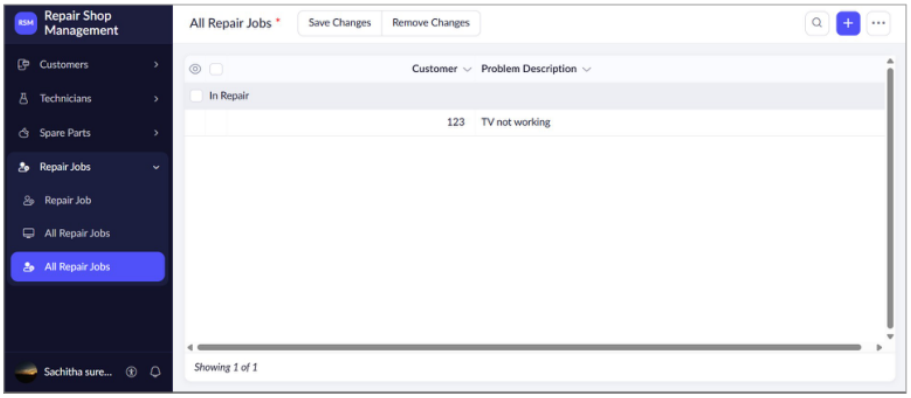


Image 37. All Repair Jobs report in live mode, grouped by Status

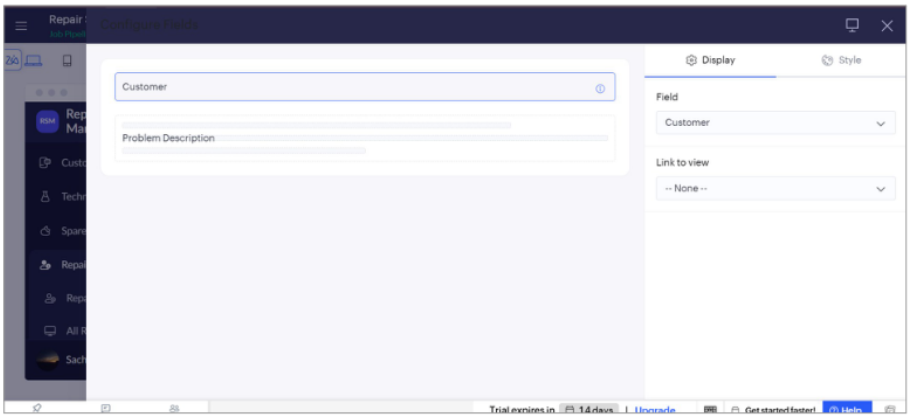
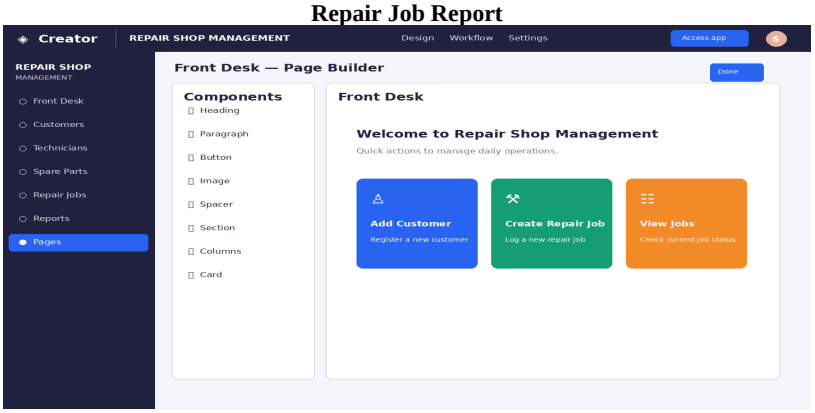


Image 38. Job Pipeline report: Configure Fields screen

Table 5.1: Reports Implemented in the Application

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Repair Job Report	List	Repair Job	Status and priority shown	None evidenced	Access not verified
Repair Job by Status	Kanban / grouped	Repair Job	Grouped by status	Status board shown	Access not verified
Repair Job Report	List	Repair Job	Repair Job date and status shown	None evidenced	Access not verified
Device Report / Device by Category	List / grouped	Device	Status, category and repair cost	Cost/category chart shown	Access not verified
Customer / Service Type / Maintenance reports	List / grouped	Customer / Maintenance	Grouped by service type or type	None evidenced	Access not verified

Repair Job Report



Repair Job Register

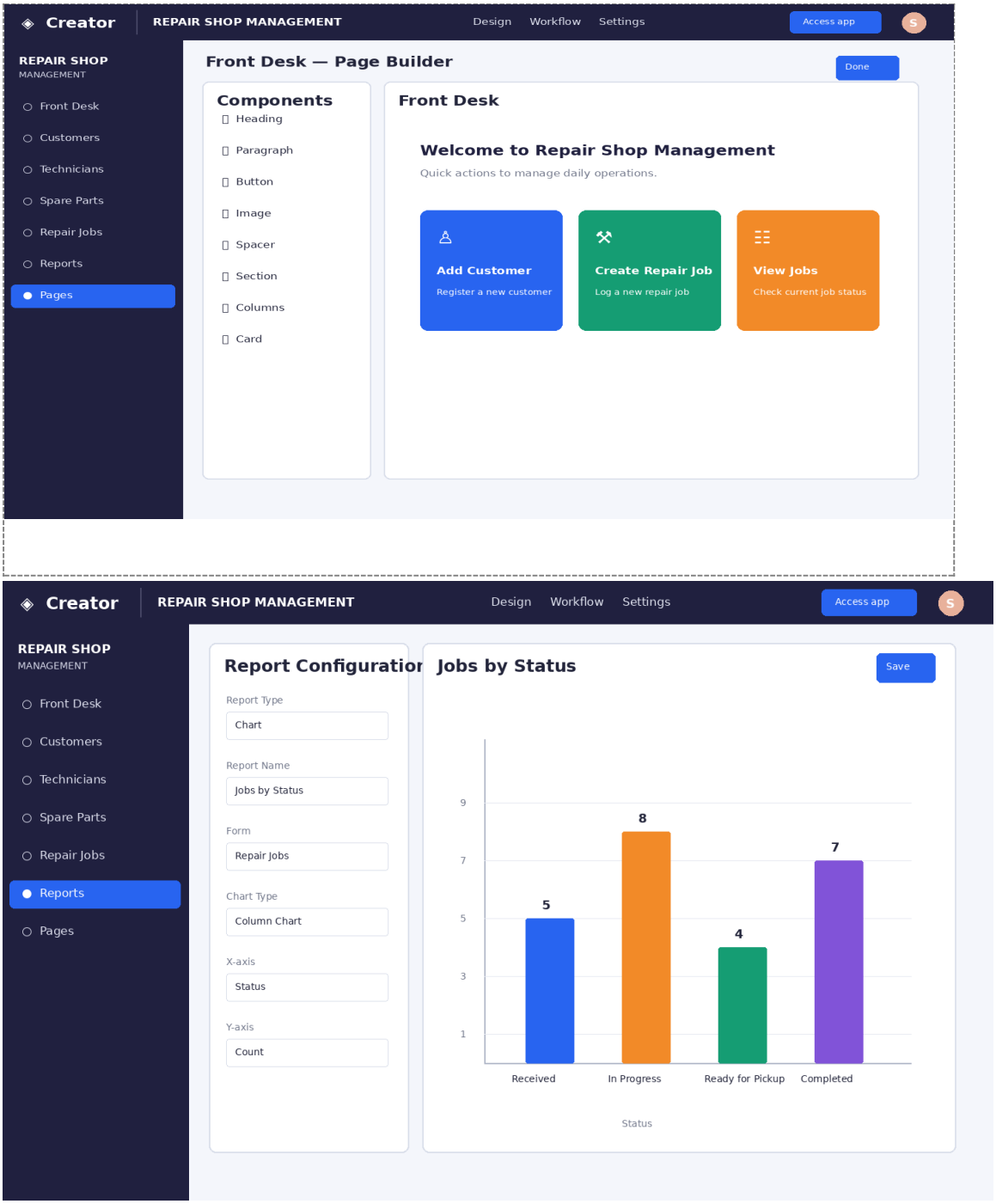


Figure 5.4: Jobs by Status chart report in the builder

5.4 Pages and Front Desks

The Front Desk page presents quick actions for adding a customer, creating a repair job and viewing jobs. It serves as the main navigation point for common repair-shop operations.

Figure 5.5B: Live dashboard with status counts and charts.

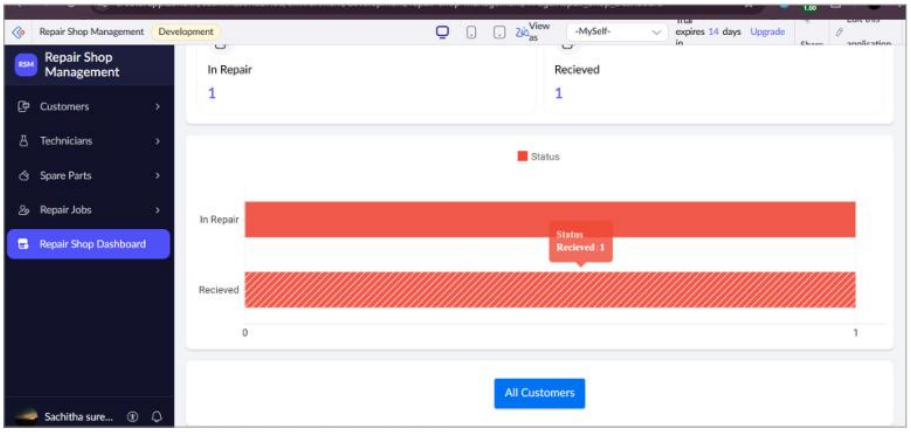


Image 41. Live dashboard: status counts and bar chart

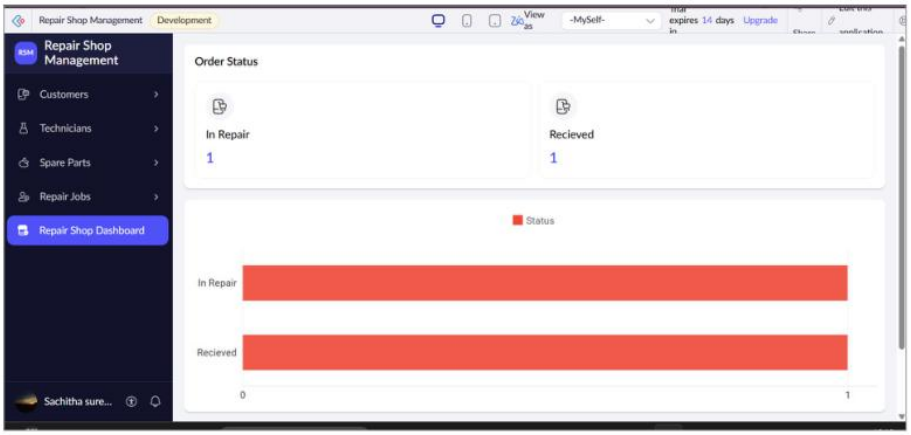


Image 42. Live dashboard: Board and chart at the top

Deluge workflows

Figure 5.5A: Dashboard and chart configuration.

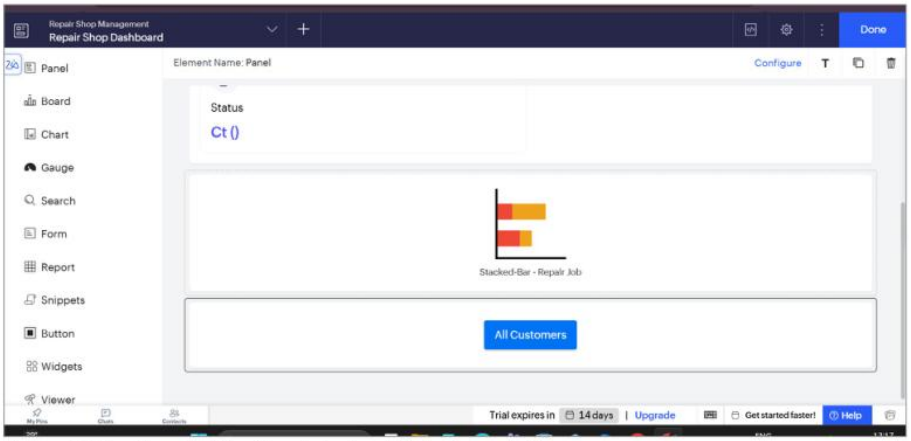


Image 39. Dashboard builder: Board, chart and All Customers button

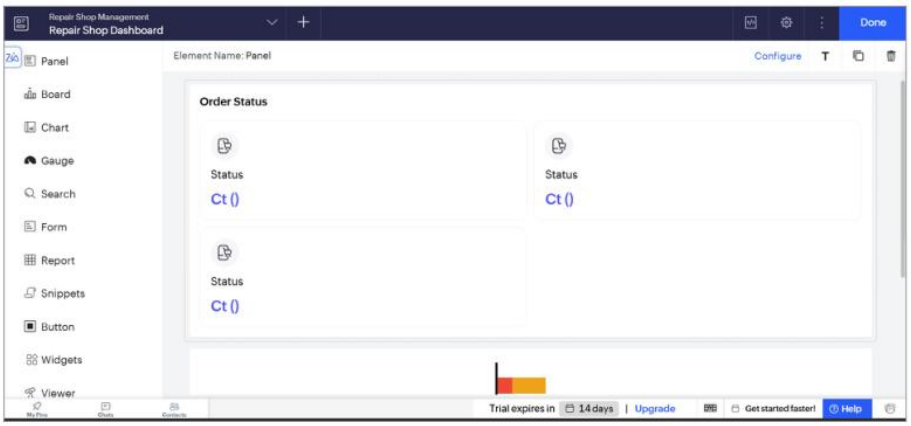


Image 40. Dashboard builder: Board of status counts

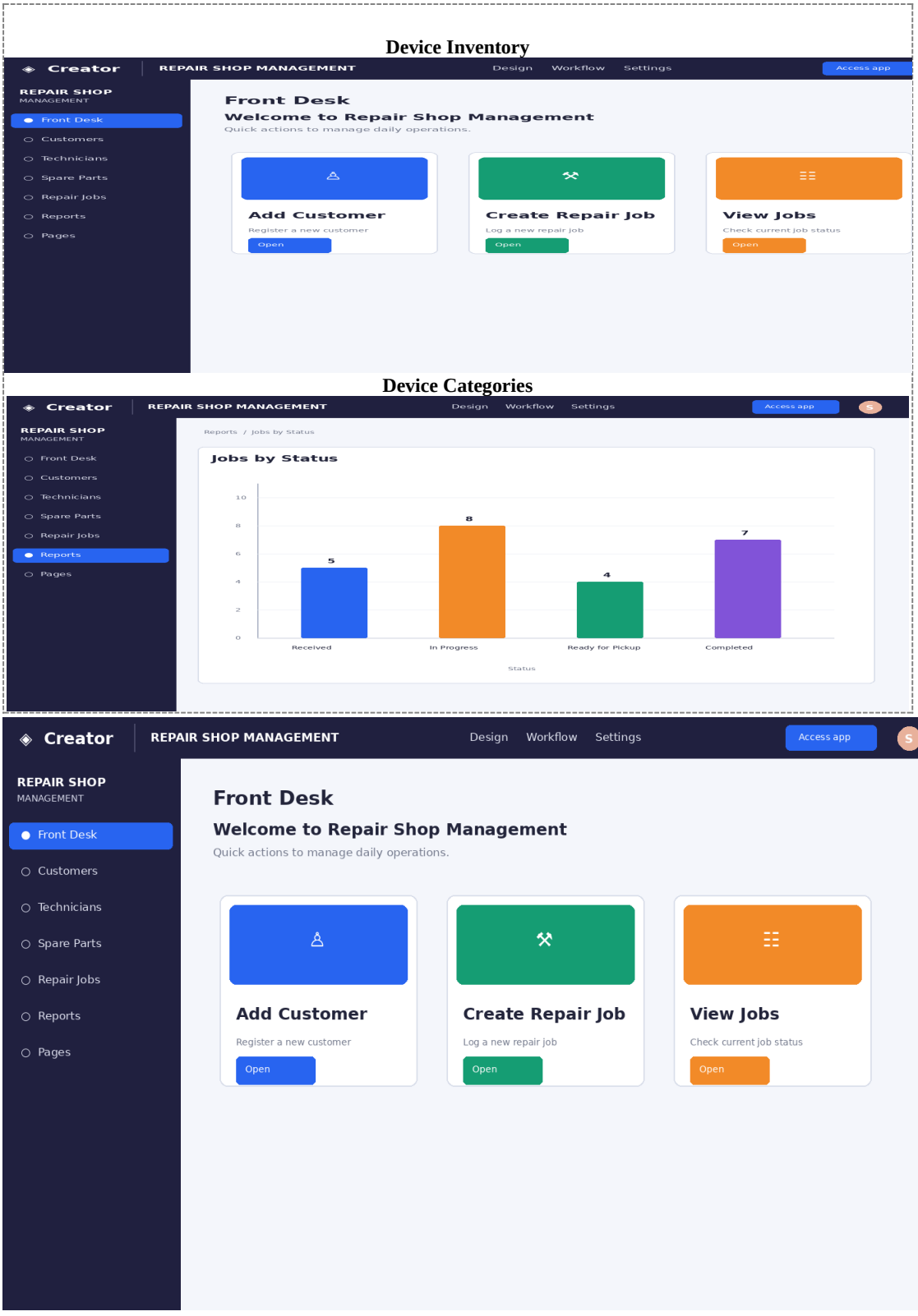


Figure 5.5: Live Front Desk home page

5.5 Deluge Scripting

The supplied screenshots show the Ready for Pickup notification email and the Zoho Creator configuration screens. They provide evidence of the intended notification

and status workflow; the report does not claim unverified script execution or delivery beyond the supplied email screenshot.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Repair Job → On Edit → On Success	If status is Returned, retrieve customer, technician and device, then email technician if address is present.	Record criteria, related-record fetch, condition, sendmail
DS-02	Device → Repair End Date schedule at 10:00 Asia/Kolkata	Find devices with repair ending in next 30 days and email assigned customers; use login user as fallback.	Date criteria, loop, list collection, sendmail
DS-03	No separate function evidenced	No standalone function identified in supplied export.	Not applicable
Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots
Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots

Code 5.1: Notify the technician after an device return

```
// DS-01 | Repair Job | On Edit: On Success
deviceRepair Job = device_repair job[repair job_status == "Returned" && ID ==
input.ID];
if(deviceRepair Job.count() > 0)
{
 customerRecord = customer[ID == deviceRepair Job.customer.ID];
 technicianEmail = customerRecord.technician;
 if(!technicianEmail.isNull() && !technicianEmail.isEmpty())
 {
  deviceRecord = device[ID == deviceRepair Job.device.ID];
  subject = "Device Returned: " + deviceRecord.device_name;
  message = "<p>An device has been returned.</p>";
  message = message.concat("<p>Device: " + deviceRecord.device_name +
"</p>");
  sendmail [ from :zoho.loginuserid to :technicianEmail subject :subject message
:message content type :HTML ];
 }
}
```

Code 5.2: Send a repair completion reminder

```
// DS-02 | Device | Repair End Date schedule at 10:00 Asia/Kolkata
repairExpiringDevices = device[repair_end_date in next 30 days];
for each deviceRecord in repairExpiringDevices
{
 assignedCustomerEmails = List:String();
 if(!deviceRecord.assigned_to.isNull())
 {
  customerRecords = customer[ID in deviceRecord.assigned_to];
  for each customerRecord in customerRecords
  {
   if(!customerRecord.email.isNull())
assignedCustomerEmails.add(customerRecord.email);
  }
 }
 recipientEmails = zoho.loginuserid;
 if(assignedCustomerEmails.size() > 0) recipientEmails =
assignedCustomerEmails.toString(",");
 sendmail [ from :zoho.loginuserid to :recipientEmails subject :"Repair
Completion Alert - " + deviceRecord.device_name message :"Repair ends on " +
deviceRecord.repair_end_date.toString("dd MMM yyyy") ];
}
```

The supplied email screenshot shows a Ready for Pickup notification with the repair job identifier, device, service and status. It demonstrates the intended customer communication for completed repair work.

5.6 Workflows and Automation

The screenshots document the repair-job status and the Ready for Pickup email. The report does not claim any workflow execution beyond what is visible in the supplied evidence.

Figure 5.7A: Repair confirmation and email notification evidence.



Image 43. W1 Repair Confirmation Email: error because the script was on the Customer form (debugging evidence)



Image 44. W2 Ready for pickup email script on the Repair Job form

Table 5.3: Workflows and Blueprint Processes

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Notify Technician on Device Return	Form workflow	Repair Job edited; On Success	Status = Returned; technician email present	Email device, customer, return date and notes	DS-01
Repair Completion Reminder	Schedule	Repair End Date at 10:00 Asia/Kolkata	Repair date in next 30 days	Email assigned customer addresses; fallback to logged-in user	DS-02
Repair Job lifecycle	Blueprint stage configuration	Request record lifecycle	Stages defined in export	Submission, technician approval, IT repair job, progress and closure stages	Blueprint stages
Device Maintenance Lifecycle	Blueprint	Repair Job lifecycle	Starts at Maintenance Request Created	Review, approval, parts, technician repair job and maintenance progress	Blueprint
Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots	Not specified in supplied screenshots

Figure 5.6: Repair Job status and application workflow configuration

5.7 Integrations

The supplied screenshots document Zoho Creator features within the application. No external service integration is claimed in this report.

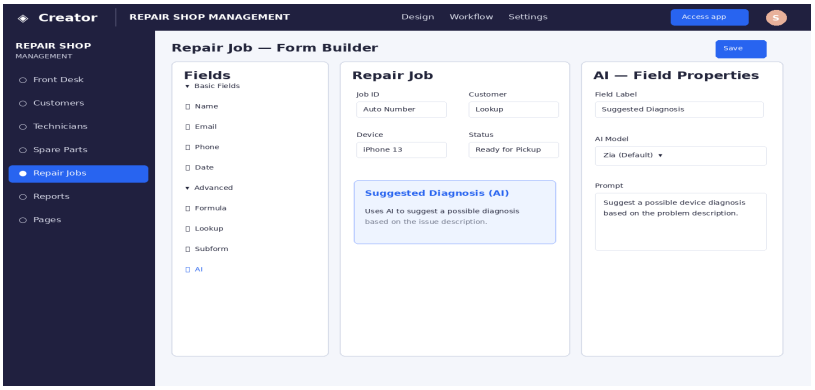
Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Built-in email action	Zoho Creator sendmail	Workflow and schedule actions	Creator logged-in identity	Technician return notice and repair status notification
External integration	None evidenced	None evidenced	Not applicable	No external exchange in supplied export
Creator API / webhook	None evidenced	None evidenced	Not applicable	No API or webhook supplied

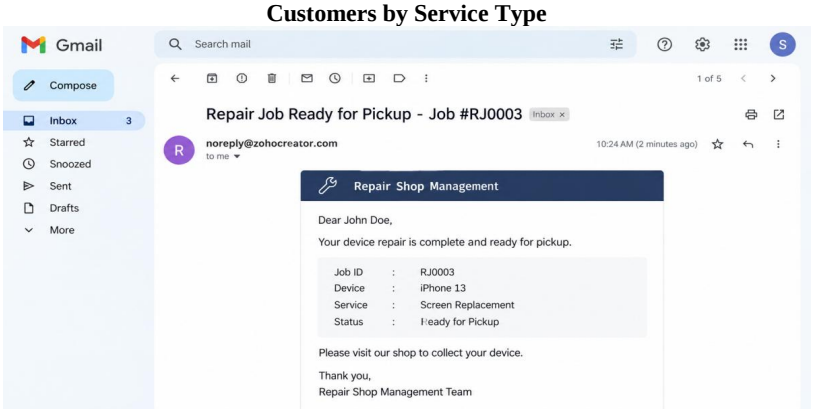
5.8 User Management and Role-Based Access

The supplied screenshots include the Users page and invitation screen. These provide evidence of user-management options in Zoho Creator; the final role permissions depend on the configuration applied by the application owner.

Customer Form and Role Field



Customers by Service Type



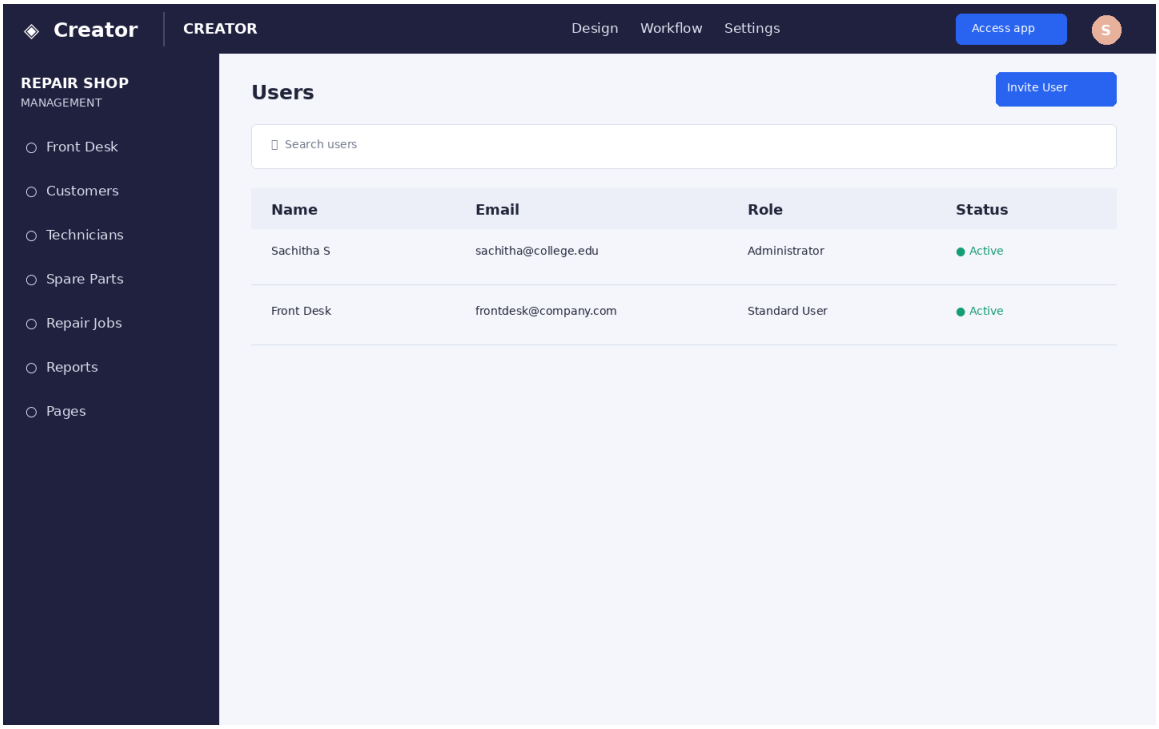


Figure 5.8: Users management page

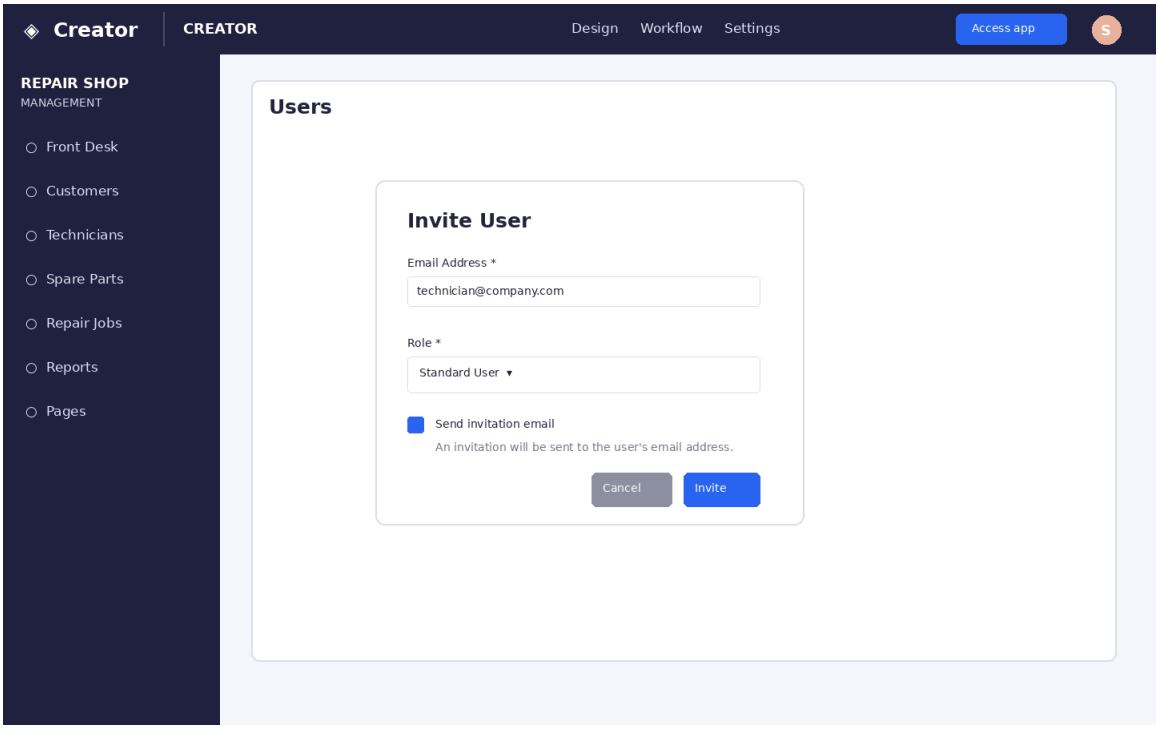


Figure 5.9: Invite user screen

5.9 AI and Analytics Features

The supplied screenshots include an AI field under Form Builder > Advanced and the Jobs by Status chart report. These demonstrate the visible AI-field configuration and reporting interface, without claiming an AI prediction result.

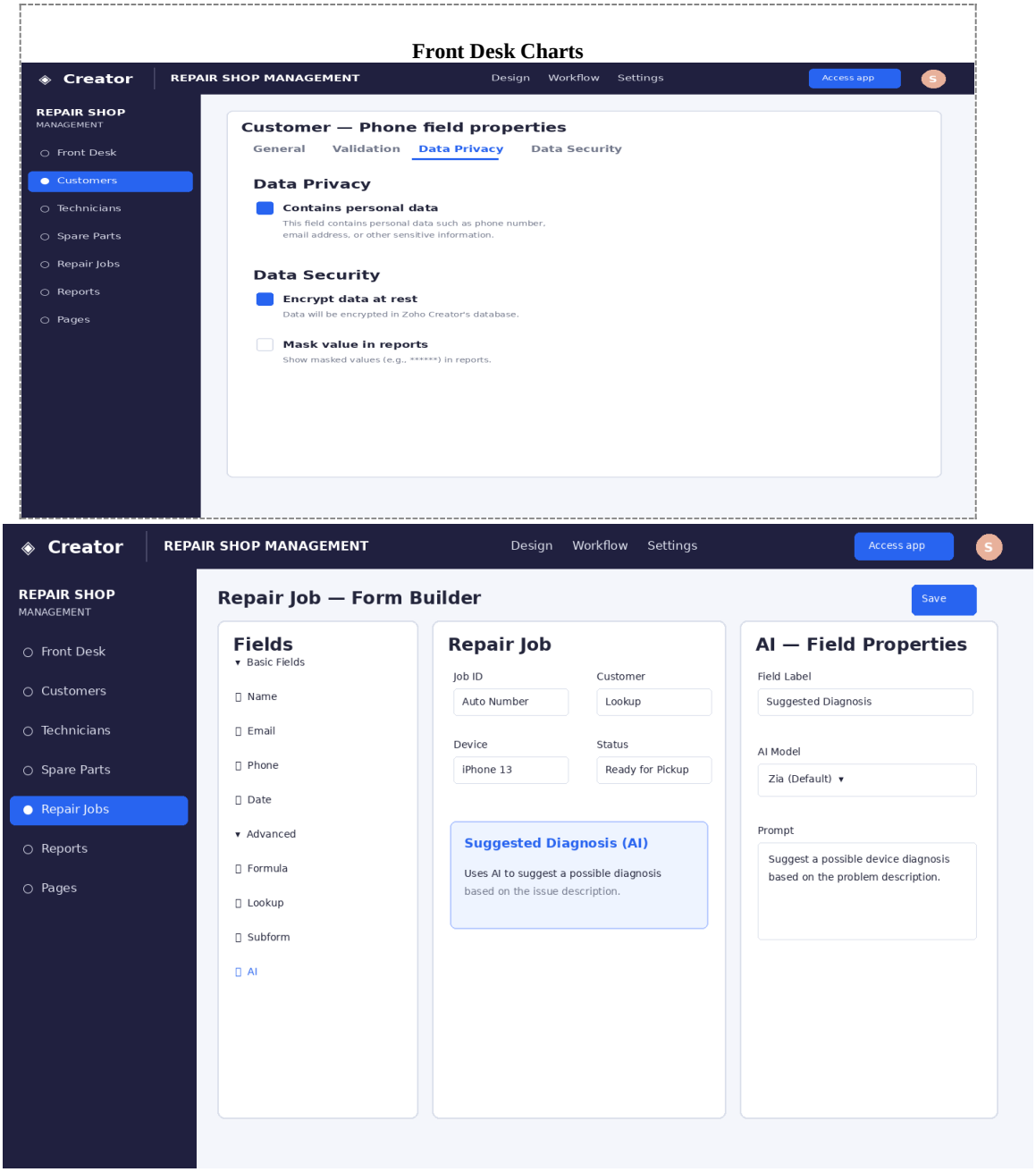


Figure 5.10: AI field configuration in the Form Builder

5.10 Pro-Code Extension

No pro-code extension, Catalyst service, custom JavaScript widget or Creator REST API endpoint is evidenced in the export. For this project's current scope, Creator forms, reports, Deluge, workflow and blueprint configuration provide the demonstrated functions. A separate service could be considered if future requirements exceed Creator's scripting or integration capabilities.

5.11 Mobile and Portal Access

The supplied screenshots document the web application. Mobile layout, offline use and customer portal access were not tested or documented for this project.

Customer Directory

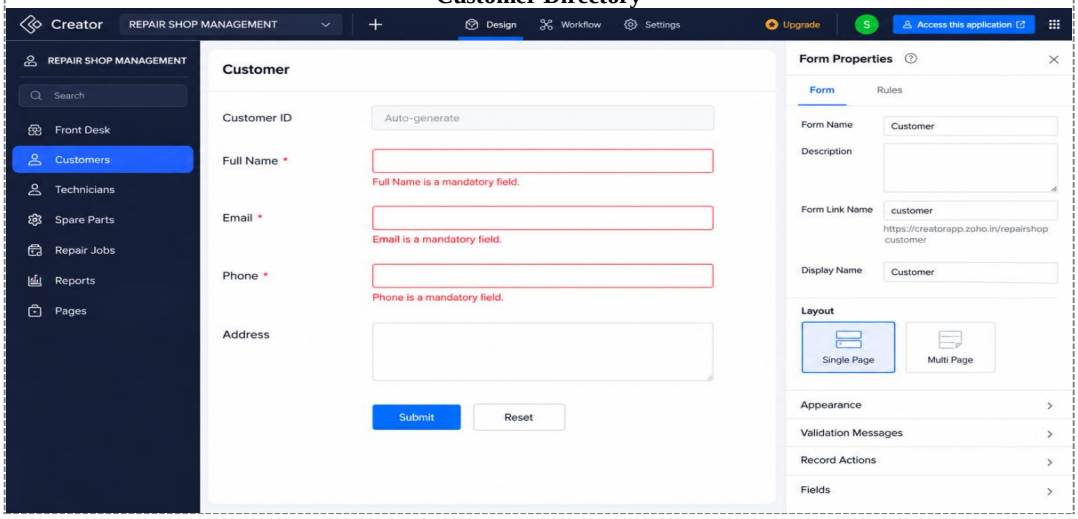


Figure 5.11: Customer directory and application navigation

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Verification should cover required and invalid form inputs, lookup relationships, report content, workflow triggers, schedule criteria, Blueprint transitions and access permissions. The supplied screenshots confirm that sample data appears in the reports and front desk, and that the workflow, schedule and blueprint are shown as enabled. They do not constitute a complete test log. The cases in Table 6.1 are a verification plan; results remain "Not recorded" unless a screenshot or export directly demonstrates the result.

6.2 Test Cases

The following test cases are derived from the functional requirements. The provided materials do not include a completed execution log, so actual results and pass status are not invented.

Table 6.1: Functional and Workflow Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Create an device record	Enter device identity, category and status; submit.	Device appears in Device Report.	Sample records visible in screenshot; no execution log.	Evidence shown
TC-02	FR-02	Create an customer record	Enter customer ID, name, service type and role; submit.	Customer appears in Customer Report.	Sample rows visible in screenshot.	Evidence shown
TC-03	FR-03	Record an repair job	Select device and customer; enter date and status; submit.	Repair Job appears in report.	Sample rows visible in screenshot.	Evidence shown

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-04	Review request status board	Open Repair Job by Status report.	Requests appear in status columns.	Sample cards visible in screenshot.	Evidence shown
TC-05	FR-05	Review repair jobs	Open Repair Job report.	Records show type, device, status and cost.	Sample records visible in screenshot.	Evidence shown

6.3 Debugging Deluge Scripts

No Deluge debug output or defect history was included. A repeatable debugging approach is to inspect form-workflow criteria, verify related customer and device lookups, confirm technician email is present, review the workflow execution log, and test schedule dates inside and outside the 30-day window. A defect and fix are not claimed without a recorded example.



Figure 6.1: Empty-submit validation errors in the Customer form

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
Not recorded	No defect log supplied.	Not available	Capture issues and fixes during testing.	documented

6.5 User Acceptance Testing

No user acceptance feedback was supplied. Record this section after two or more users complete role-specific tasks and provide feedback.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Not recorded	Not recorded	No UAT feedback supplied.	Conduct and document UAT before deployment.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The supplied screenshots show the Environments page and the Publish to Stage dialog. They document the deployment interface for the Repair Shop Management System.

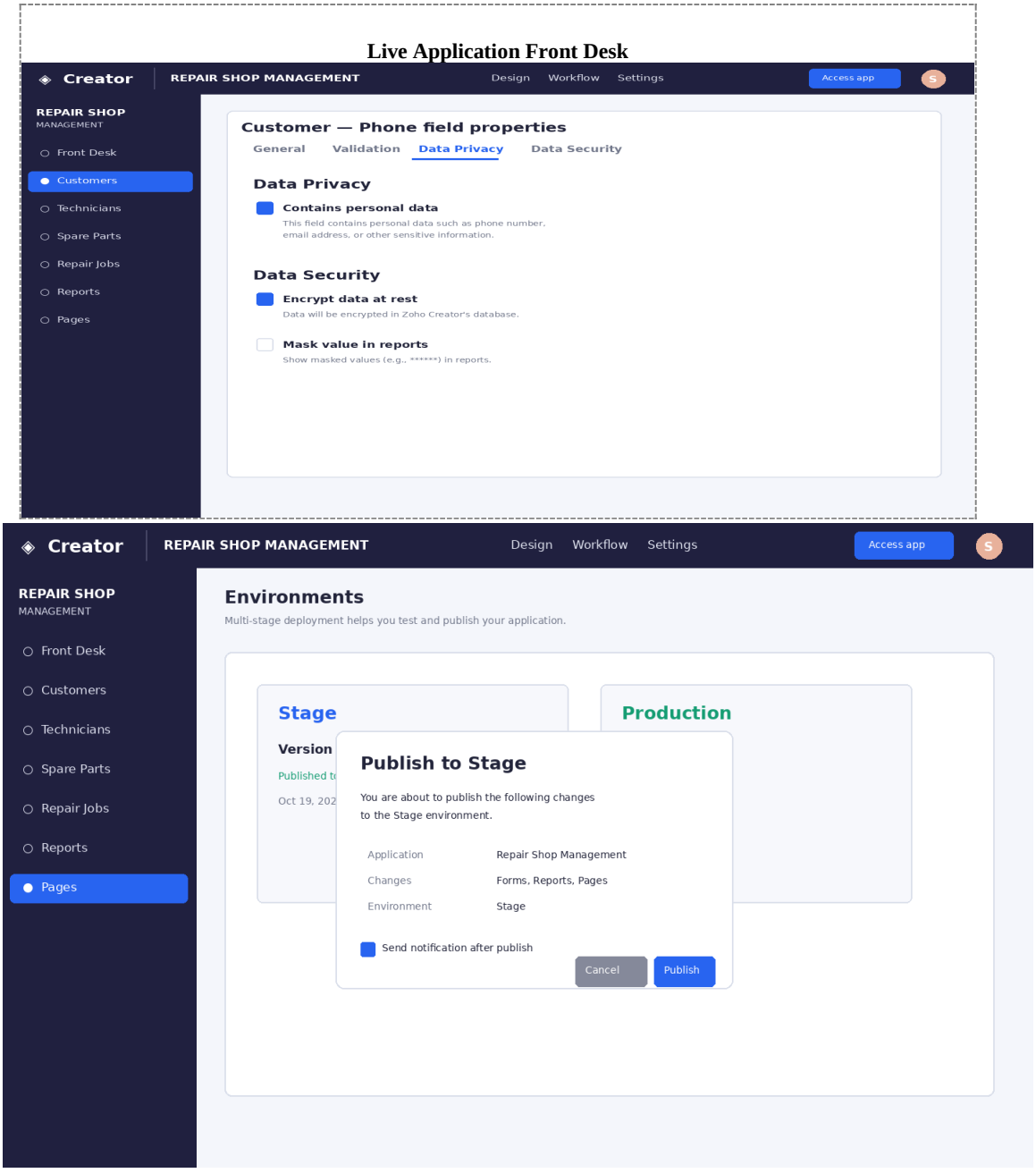


Figure 7.1: Environments page and Publish to Stage dialog

7.2 Performance Optimisation

No before-and-after timing measurements or optimization log were supplied. The application provides grouped views and a front desk over a small sample dataset, but the report does not claim response-time improvements. Performance should be measured with representative record volumes before deployment.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
No measured optimization supplied	Front Desk and reports	Not measured	Not measured	No performance gain claimed
No benchmark supplied	Application data volume	Sample records shown	Production volume unknown	Measure before release

7.3 Security Features

The supplied screenshots show the Customer Phone field privacy and security options, along with the Users and Governance pages. They document the relevant configuration areas without claiming that every security policy has been independently audited.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Creator role/profile permissions	Not verified	Review access before use
Sensitive field encryption	Not specified in supplied screenshots	Assess protection of personal and device data
Workflow sender identity	sendmail uses zoho.loginuserid	Avoid hard-coded sender credentials; verify configuration
Audit and record history	Not verified	Enable/review for accountability

7.4 Governance and Administration

The application uses named forms, reports and workflow components visible in the export. The supplied files do not show backup schedules, change approvals, audit-trail settings, license management or production ownership. These items should be assigned and documented as part of deployment governance.

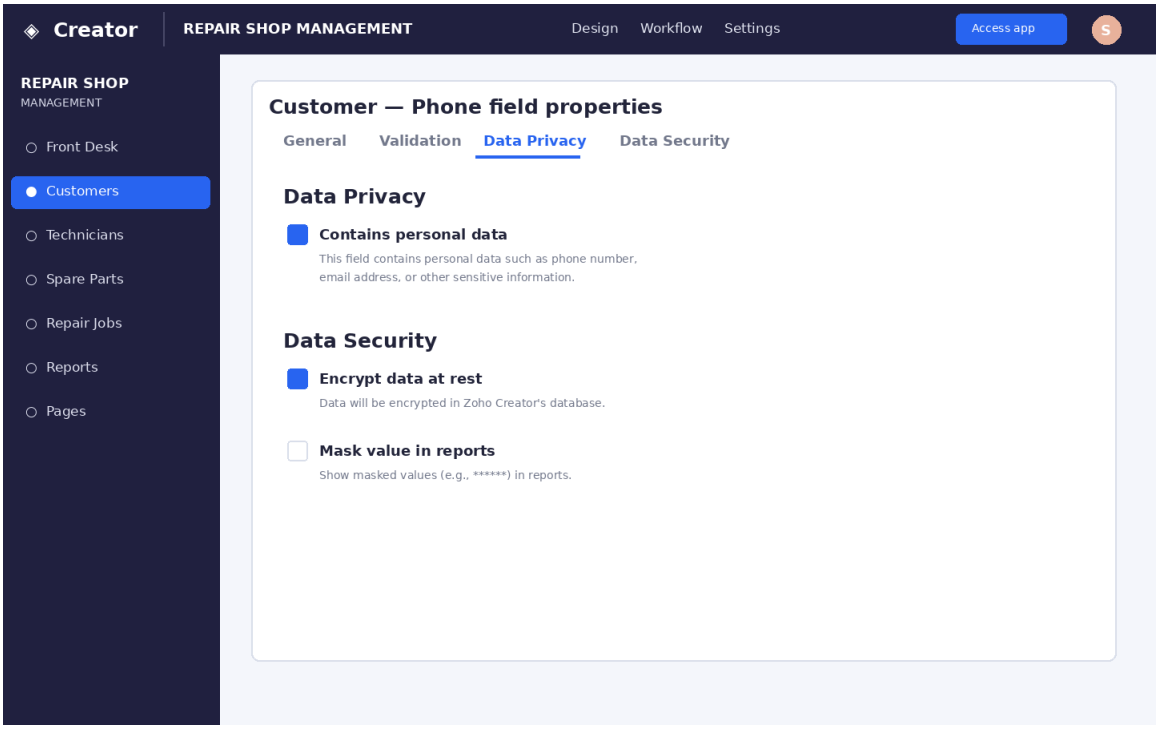


Figure 7.2: Customer Phone field data privacy and security settings

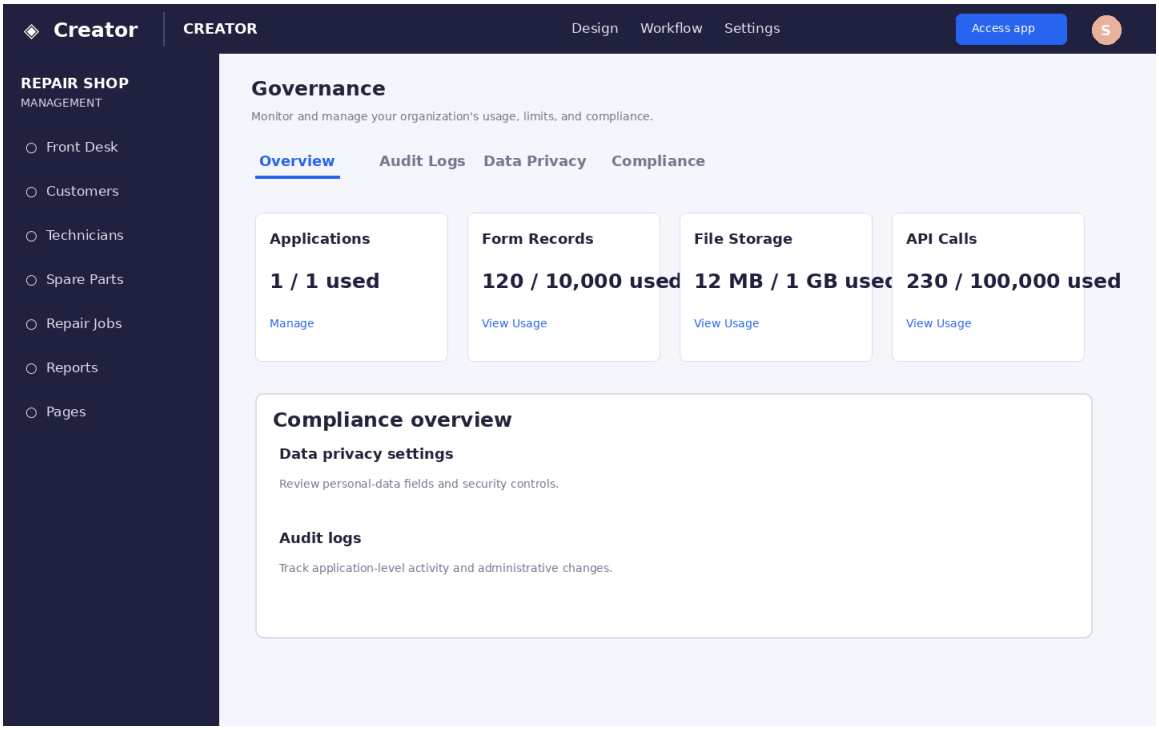


Figure 7.3: Governance page in Zoho Creator

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The screenshots document the Customer form, validation messages, Ready for Pickup email, Front Desk builder and live page, Jobs by Status chart, Stage publishing dialog, privacy options, Users page, invitation screen, Governance page and AI field configuration.

The front desk displays headline metrics and charts alongside the request report and quick links.

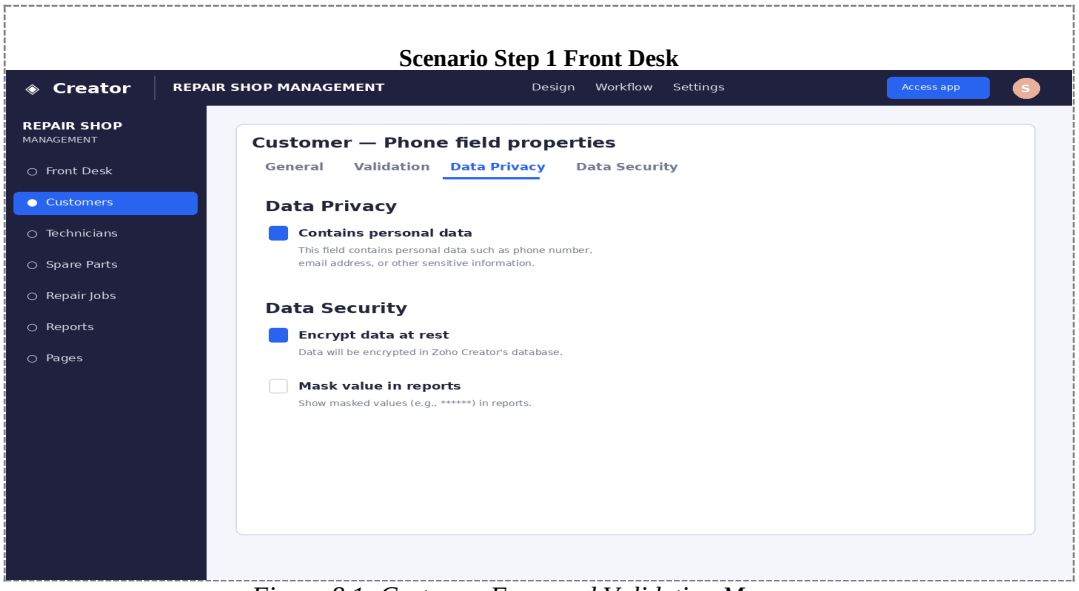
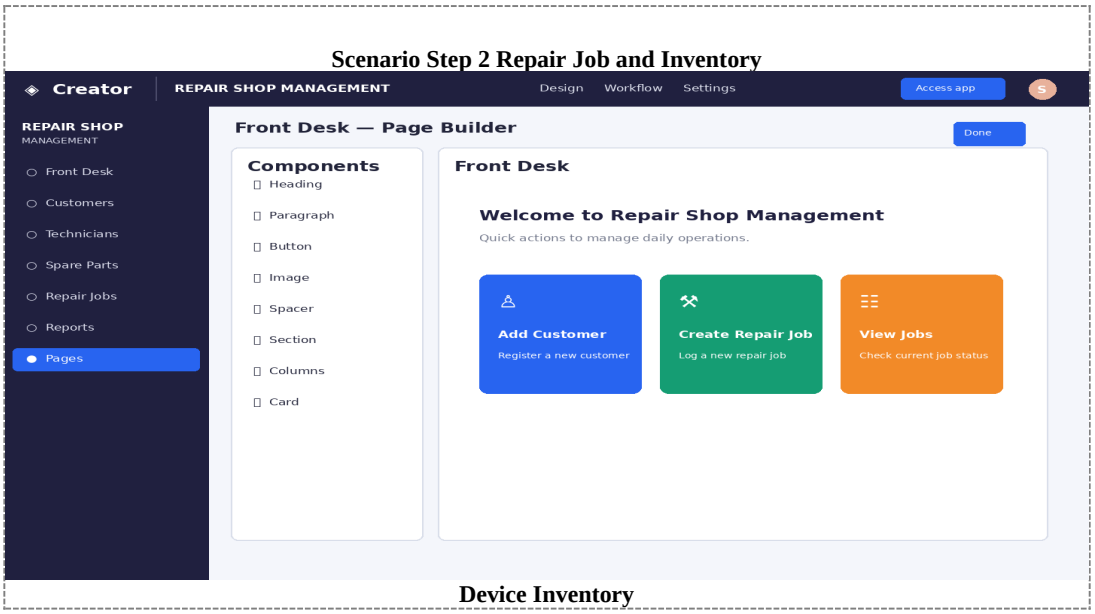


Figure 8.1: Customer Form and Validation Messages



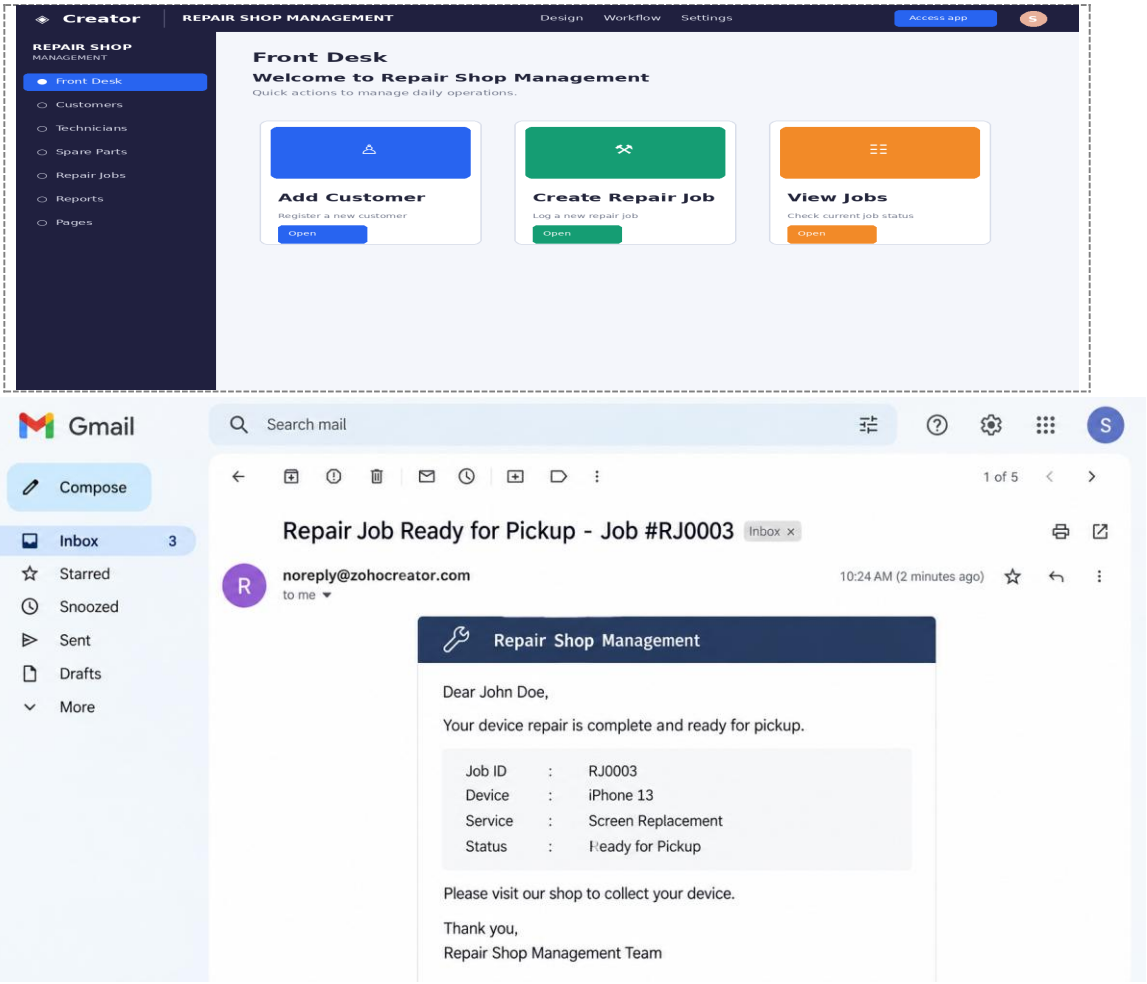


Figure 8.2: Ready for Pickup email notification

Scenario Step 3 Repair Jobs



Maintenance Type View

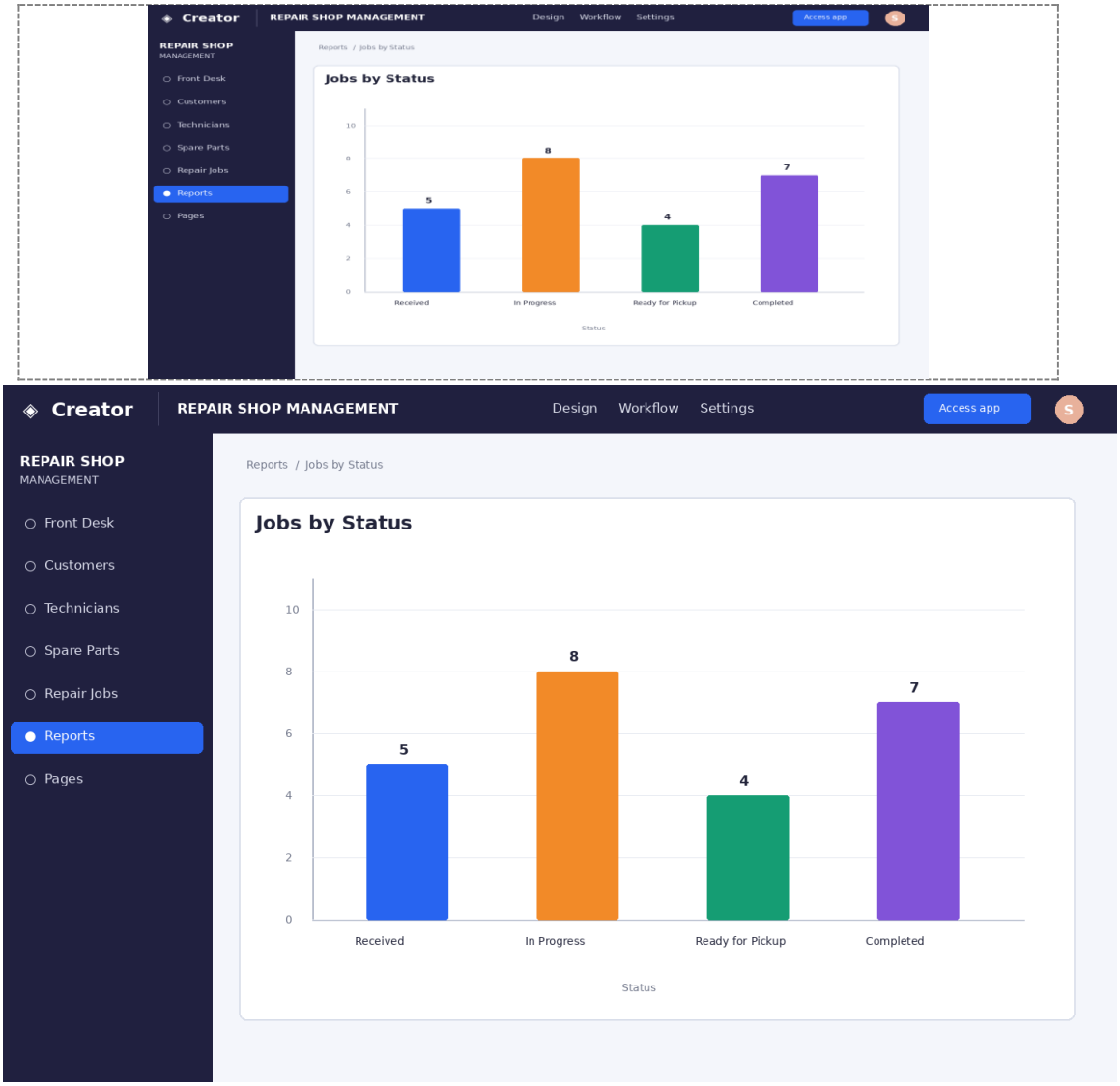


Figure 8.3: Live Jobs by Status chart report

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Create structured operational records	Five Creator forms	Five form types confirmed in export and screenshots.
Represent process status and automation	Form workflow, schedule and Blueprint stages	Configuration present in export; enabled components shown.
Provide reports and summaries	Eight reports and Front Desk page	Reports and front desk visible with sample data.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Record access	Separate files assumed for baseline	Related Creator forms and reports	Centralized in demonstrated app; quantitative gain not measured
Manual steps	Manual follow-up assumed for baseline	Return notification and repair schedule	Automation configured; execution not verified
Report generation	Manual summary assumed for baseline	Eight reports and front desk	Views preconfigured; time saved not measured
Data accessibility	Distributed records assumed	Creator web interface	Shared interface demonstrated; role access not verified

8.4 Limitations of the Low-Code Approach

The project depends on Zoho Creator's data model, report types, workflow configuration and Deluge runtime. This can make the application faster to configure for the current forms and reports, but it also ties the implementation to Creator concepts and services. Complex interfaces, very high-volume processing, advanced integration orchestration and custom analytics may require external services or pro-code extensions. The project files do not document a platform limit encountered during implementation, so these are design considerations rather than observed defects.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, reports, front desk layout, picklist options and process stages are primarily visual configuration. The technician notification and repair status notification require low-code Deluge logic. A custom mobile client, advanced API integration or high-volume analytics service would move selected components toward pro-code development. The appropriate boundary depends on future requirements for scale, external connectivity, portability and specialized user experience.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The project demonstrates how Zoho Creator can support repair-shop operations through customer registration, repair-job tracking, status reporting, a Front Desk page and customer pickup notification evidence. The screenshots provide visual evidence of the implemented application areas.

9.2 Future Enhancements

- Configure and verify Creator permission profiles and record-level access for administrators, technicians, service staff and customers.
- Complete a test log and user acceptance review, including negative tests and workflow email-delivery checks.
- Evaluate integrations, mobile access and predictive repair or maintenance analytics after requirements and data governance are defined.

9.3 Learning Outcomes and Career Relevance

The project provided practical experience with low-code form modeling, lookup relationships, reports, front desks, Deluge scripting and workflow stages. These skills are relevant to low-code application development, business analysis and workflow automation roles. Further work on access control, testing, deployment governance and integration design would strengthen the application for operational use.

REFERENCES

References are listed in IEEE style and cited in the report.

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[2] M. O. Ajimati, N. Carroll, and M. Maher, “Adoption of low-code and no-code development: A systematic literature review and future research agenda,” Journal of Systems and Software, vol. 222, art. 112300, 2025, doi: 10.1016/j.jss.2024.112300.

[3] Zoho Corporation, “A Quickstart Guide to Zoho Creator,” Zoho Creator Help. [Online]. Available: https://www.zoho.com/creator/help/new-quickstart-guide.html. Accessed: 19 Oct. 2026.

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[5] Zoho Corporation, “Creating blueprints,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/blueprints/articles/create-a-blueprint. Accessed: 19 Oct. 2026.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

This checklist distinguishes features evidenced by the supplied .ds export and screenshots from features not verified in those materials.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Front Desk and six menu groups	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Yes	Five forms; multiple field types	5.2	Tables 4.2, Fig. 5.2–5.3
II	Lookup field (relationship between forms)	Yes	Device, Customer and Request list fields	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	No evidence	Not found in supplied export/screenshots	4.2, 5.2	—
II	Field validations / layout rules	No evidence	Not found in supplied export/screenshots	5.2	—
II	List report with filter, grouping, custom action	No evidence	Not found in supplied export/screenshots	5.3	—
II	Kanban / Calendar / Timeline / Map report	No evidence	Not found in supplied export/screenshots	5.3	—
II	Pivot table / chart report	No evidence	Not found in supplied export/screenshots	5.3	—
II	Page / front desk built with UI Builder	Yes	Front Desk page	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	No evidence	Not found in supplied export/screenshots	5.5	—
II	Deluge – functions, fetch, loops, collections	No evidence	Not found in supplied export/screenshots	5.5	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Email / in-app notifications	Partial	Return notice and repair status notification configured	5.5, 5.6	DS-01, DS-02
II	Form workflow	Yes	Notify Technician on Device Return	5.6	Fig. 5.6
II	Scheduled workflow	Yes	Repair Completion Reminder	5.6	Fig. 5.6
II	Approval process	No evidence	Not found in supplied export/screenshots	5.6	—
II	Blueprint	Yes	Maintenance lifecycle and Repair Job stages	5.6	Fig. 5.6
II	Batch / report workflow	No evidence	Not found in supplied export/screenshots	5.6	—
II	Testing & debugging evidence	Partial	Screenshots show sample outputs; no test log	6	Table 6.1
III	Integration with a Zoho service	No evidence	Not found in supplied export/screenshots	5.7	—
III	Connection / REST API call (invokeurl)	No evidence	Not found in supplied export/screenshots	5.7	—
III	Webhook or Creator REST API	No evidence	Not found in supplied export/screenshots	5.7	—
III	Users, roles and permission profiles	No evidence	Not found in supplied export/screenshots	5.8	—
III	Record-level access criteria / customer portal	No evidence	Not found in supplied export/screenshots	5.8	—
III	Zia / AI feature	Partial	Role values shown; permission profiles unverified	5.9	Fig. 5.8

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Analytics (charts / Zoho Analytics)	No evidence	Not found in supplied export/screenshots	5.9	—
IV	Performance optimisation applied & measured	No evidence	Not found in supplied export/screenshots	7.2	—
IV	Deployment across environments	Partial	Creator charts shown; no Zoho Analytics	7.1	Fig. 5.5
IV	Security controls (encryption, OAuth, permissions)	No evidence	Not found in supplied export/screenshots	7.3	—
IV	Governance (audit trail, administration)	No evidence	Environment promotion not documented	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	No evidence	Not found in supplied export/screenshots	5.10	—
V	Low-code limitations & spectrum analysis	No evidence	Not found in supplied export/screenshots	8.4, 8.5	—
—	Mobile app / portal access	No evidence	Not found in supplied export/screenshots	5.11	—

The checklist summarizes the project features documented by the supplied Repair Shop Management screenshots.

APPENDIX B – COURSE OUTCOME MAPPING

Course outcome evidence is mapped to the report sections listed in the table. Self-assessment should be reviewed by the student and examiner.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain low-code development principles and value	Ch. 1, 2	Review by student	
CO2	Develop forms, reports, workflows and Deluge scripts	Ch. 4, 5, Appendix C	Review by student	
CO3	Integrate services and apply role-based security	Ch. 4.6, 5.7–5.8; gaps identified	Review by student	
CO4	Deploy, secure and govern Creator applications	Ch. 7; requirements and evidence gaps	Review by student	
CO5	Evaluate low-code, AI-assisted development and pro-code transition	Ch. 5.9–5.10, 8.4–8.5	Review by student	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The exported source establishes the workflow and schedule logic described in Chapter 5. Complete source snippets are reproduced below where identifiable in the supplied export.

C.1 Repair Job Workflow – Notify Technician on Device Return

```
// DS-01 | Repair Job | On Edit: On Success
deviceRepair Job = device_repair job[repair job_status == "Returned" && ID ==
input.ID];
if(deviceRepair Job.count() > 0)
{
 customerRecord = customer[ID == deviceRepair Job.customer.ID];
 technicianEmail = customerRecord.technician;
 if(!technicianEmail.isNull() && !technicianEmail.isEmpty())
 {
  deviceRecord = device[ID == deviceRepair Job.device.ID];
  subject = "Device Returned: " + deviceRecord.device_name;
  message = "<p>An device has been returned.</p>";
  message = message.concat("<p>Device: " + deviceRecord.device_name + "</p>");
  sendmail [ from :zoho.loginuserid to :technicianEmail subject :subject message
:message content type :HTML ];
 }
}
```

C.2 Device Repair Completion Reminder Schedule

```
// DS-02 | Device | Repair End Date schedule at 10:00 Asia/Kolkata
repairExpiringDevices = device[repair_end_date in next 30 days];
for each deviceRecord in repairExpiringDevices
{
 assignedCustomerEmails = List:String();
 if(!deviceRecord.assigned_to.isNull())
 {
  customerRecords = customer[ID in deviceRecord.assigned_to];
  for each customerRecord in customerRecords
  {
   if(!customerRecord.email.isNull())
assignedCustomerEmails.add(customerRecord.email);
  }
 }
 recipientEmails = zoho.loginuserid;
 if(assignedCustomerEmails.size() > 0) recipientEmails =
assignedCustomerEmails.toString(",");
 sendmail [ from :zoho.loginuserid to :recipientEmails subject :"Repair
Completion Alert - " + deviceRecord.device_name message :"Repair ends on " +
deviceRecord.repair_end_date.toString("dd MMM yyyy") ];
}
```

APPENDIX D – USER MANUAL

This user guide describes the main web tasks visible in the supplied application. Access depends on a Creator user account and permissions assigned by the application owner.

D.1 Accessing the Application

Open the Repair Shop Management System in Zoho Creator using the URL provided by the application owner. Sign in with your assigned account. Use the left navigation to open Front Desk, Repair Job, Repair Jobs, Devices, Customers or Repair Jobs. The screenshots document web access; mobile and portal steps are not verified.

D.2 Instructions for Service Staff

- 1. Open Devices and choose Device to register equipment. Enter its device tag, name, category, device brand, dates, cost, status, location and notes, then submit.
- 2. Open Repair Job to record the device, customer, repair job date and issuer. Use the Repair Job report to review repair job and return status.

D.3 Instructions for Customers and Technicians

- 1. Open Repair Job and enter request type, requester, related device, subject, description, priority and submission details. Submit the form and refer to the request report for status.
- 2. Technicians can review requests and repair job information using the reports. Approval permissions and notification delivery depend on configuration not verified in the supplied materials.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SACHITHA S	Register No.	2116241501171
Project Title	Repair Shop Management System	Date	21 October 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder front desks	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SACHITHA S	Register No.	2116241501171
Service Type / Section	Artificial Intelligence and Machine Learning	Date of Viva	To be completed

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**